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Refining the chronostratigraphy of the Karoo Basin, South Africa: magnetostratigraphic constraints support an early Permian age for the Ecca Group

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SUMMARY

The Beaufort Group of the Karoo Basin, South Africa provides an important chrono- and biostratigraphic record of vertebrate turnovers that have been attributed to the end-Permian mass extinction events at *ca.* 252 and 260 Ma. However, an unresolved controversy exists over the age of the Beaufort Group due to a large data set of published U–Pb SHRIMP (Sensitive High Resolution Ion Microprobe) zircon results that indicate a *ca.* 274–250 Ma age range for deposition of the underlying Ecca Group. This age range requires the application of a highly diachronous sedimentation model to the Karoo Basin stratigraphy and is not supported by published palaeontologic and palynologic data. This study tested the strength of these U–Pb isotopic data sets using a magnetostratigraphic approach. Here, we present a composite ∼1500 m section through a large part of the Ecca Group from the Tanqua depocentre, located in the southwestern segment of the Karoo Basin. After the removal of two normal polarity overprints, a likely primary magnetic signal was isolated at temperatures above 450 °C. This section is restricted to a reverse polarity, indicating that it formed during the Kiaman Reverse Superchron (*ca.* 318–265 Ma), a distinctive magnetostratigraphic marker for early−middle Permian rocks. The Ecca Group has a corresponding palaeomagnetic pole at 40.8°S, 77.4°E (A_{95} = 5.5°). U–Pb SHRIMP ages on zircons are presented here for comparison with prior isotopic studies of the Ecca Group. A weighted mean U–Pb age of 269.5 ± 1.2 Ma was determined from a volcanic ash bed located in the uppermost Tierberg Formation sampled from the O + R1 research core. The age is interpreted here as a minimum constraint due to a proposed Pb-loss event that has likely influenced a number of published results. A comparison with the Geomagnetic Polarity Time Scale as well as published U–Pb TIMS ages from the overlying Beaufort Group supports a *ca.* 290–265 Ma age for deposition of the Ecca Group.

Key words: Magnetic fabrics and anisotropy; Magnetic field variations through time; Magnetostratigraphy; Palaeomagnetism; Reversals: process, time scale, magnetostratigraphy; Sedimentary basin processes.

1 INTRODUCTION

The Karoo Basin of South Africa (Fig. 1) represents an extensive depositional record of Pennsylvanian–early Jurassic fossiliferous sedimentary (Rubidge *et al.* 1999) rocks that have been critical for understanding the rate of vertebrate turnovers for continental extinction events during the end-Guadalupian and across the Permian–Triassic boundary (PTB, Smith & Ward 2001; Smith & Botha 2005; Ward *et al.* 2005; Botha & Smith 2006; Rubidge *et al.* 2013; Smith & Botha-Brink 2014; Day *et al.* 2015; Gastaldo *et al.* 2015). The excellent preservation of the associated vertebrate assemblages has led to a large library of published faunal information,

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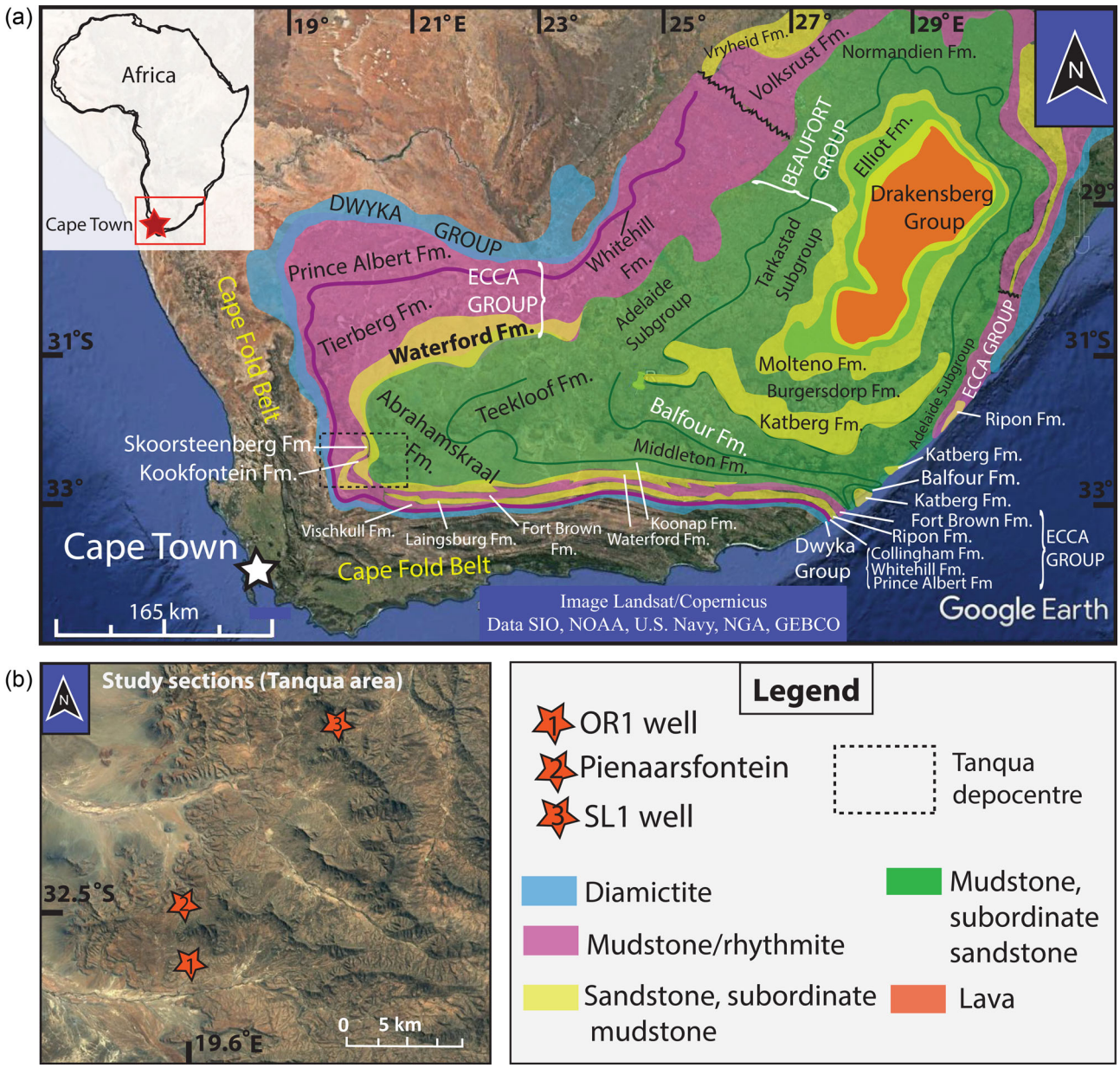


Figure 1. (a) Geological map of the Karoo Basin after Johnson *et al.* (1996). (b) Regional inset of the sampling area located in the Tanqua depocentre. Sample sections from this study are shown as red stars. For a general lithostratigraphic summary, see Fig. 2.

complemented by palynological data that has been extrapolated globally to understand the patterns of tetrapod extinctions and their timing relative to the end-Permian marine extinction at *ca.* 252 Ma (Mundil *et al.* 2004; Ward *et al.* 2005; Lucas 2010; Shen *et al.* 2011; Burgess *et al.* 2014).

The PTB is defined by marine conodont assemblages, and has a corresponding age of 251.9 ± 0.0 Ma determined from high-resolution U–Pb TIMS (zircon) dating of ash beds from the Meishan section in south China (Burgess *et al.* 2014). The terrestrial PTB is considered to be broadly represented by the Last Appearance Datum of the late Permian tetrapods of the *Daptocephalus* (formerly *Dicynodon*; Viglietti *et al.* 2016) Assemblage Zone and the First Appearance Datum (FAD) of early Triassic taxa in the *Lystrosaurus* Assemblage Zone (Lucas 2010; Smith & Botha-Brink 2014), of which the Beaufort Group of the Karoo Basin has long been considered the prime example. However, in the past decade a number of U–Pb SHRIMP (Sensitive High Resolution Ion Microprobe, zircon) results have been published from these regions that support a *ca.* 274–250 Ma age range for deposition of the underlying Ecca Group, and therefore a Triassic age for the Beaufort Group (Fig. 2, Fildani *et al.* 2007, 2009; McKay *et al.* 2015, 2016).

This depositional age assignment requires that the PTB be placed within the upper Skoorsteenberg Formation of the Ecca Group, ~2000 m below the biostratigraphically constrained PTB in the upper Beaufort Group (Fig. 2). This assignment is in direct conflict with an extensive body of palaeontologic data (Rubidge 1990; Smith & Keyser 1995; Rubidge *et al.* 1999; Barbolini 2014), as well as a series of *ca.* 262–268 Ma U–Pb SHRIMP ages on zircon grains from the overlying Abrahamskraal Formation of the lower Beaufort Group (Lanci *et al.* 2013) and *ca.* 262–255 Ma ages from the Adelaide Subgroup of the Beaufort Group (Rubidge *et al.* 2013). The apparent age controversy was explored by McKay *et al.* (2015, 2016) who demonstrated that volcanic tuffs from the Ecca Group yield consistently younger ages (250–274 Ma) than tuffs from

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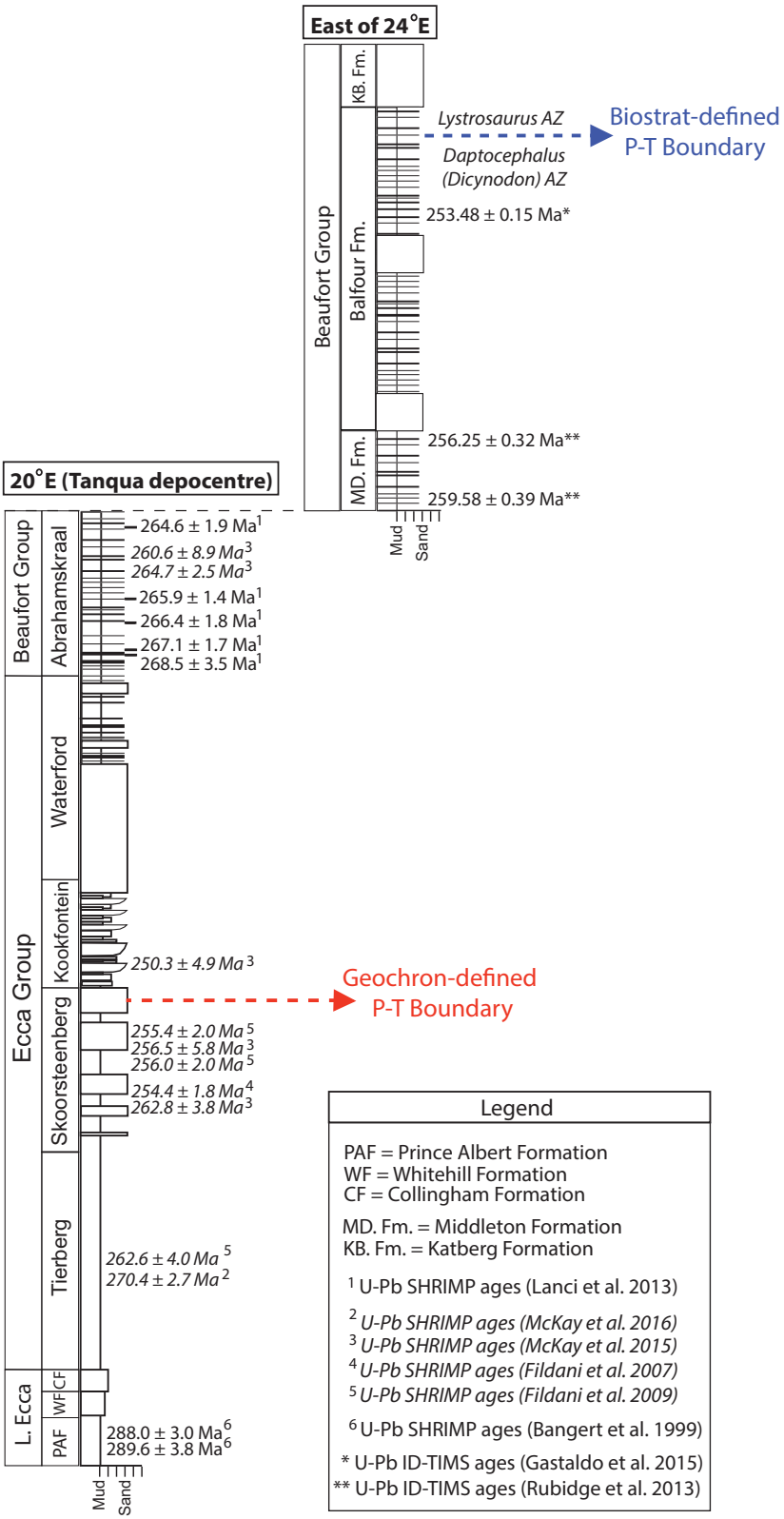


Figure 2. Lithostratigraphy and published U–Pb ages for the Permian Ecca Group of the Tanqua depocentre (20°E) and the Permian–Triassic Beaufort Group vertebrate type section (~26°E). Biostratigraphy after Smith & Botha-Brink (2014) and Gastaldo *et al.* (2017). Note the large stratigraphic inconsistency between the reported biostratigraphy and U–Pb SHRIMP ages.

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the conformably overlying Beaufort Group (257–452 Ma) along a 650 km transect across the southern margin of the Karoo Basin. This 'inverted' basin stratigraphy was ascribed to an episode of zircon exhaustion and magmatic recycling (as opposed to a disturbed U–Pb system). The placement of the PTB within the upper Ecca Group was further reinforced by a presumed 'tuff gap' in the Karoo Basin located above this horizon (Waterford Formation), similar to other Permian−Triassic depositional sites along the southern Panthalassan margin of Gondwana (Veevers 2004; McKay *et al.* 2015, 2016).

In order to resolve this apparent discrepancy between the biostratigraphically constrained and geochronologically constrained PTB, we applied the technique of magnetostratigraphy, which has proven very useful in the correlation and timing of major geological events (Steiner *et al.* 1989; Opdyke & Channell 1996; Opdyke *et al.* 2000; Nawrocki 2004; Ward *et al.* 2005; Steiner 2006; Glen *et al.* 2009; Taylor *et al.* 2009; Horacek *et al.* 2010; Szurlies *et al.* 2012; Szurlies 2013; Ogg *et al.* 2016; Hounslow *et al.* 2017). Magnetostratigraphy is a geophysical relative-dating method that relies on the global synchronicity of geomagnetic reversals (Opdyke & Channell 1996). Sedimentary rocks may record a depositional remanent magnetization (DRM) that preserves the polarity and direction of Earth's magnetic field during the time of formation (Butler 1992). The Geomagnetic Polarity Time Scale (GPTS) is the record of reversals tied to the biostratigraphically constrained geological stage boundaries (Fig. 3). The magnetic field reverses its polarity on average every 0.1–1 Ma, but reversal rates through time can be highly variable. Several periods of stable field polarity have been observed, the longest of which is known as the Kiaman Reverse Superchron (KRS; Fig. 3), which lasted more than 50 Myr (Irving & Parry 1963; Gose & Helsley 1972; Diehl & Shive 1979, 1981; Khramov & Tarling 1987; McFadden *et al.* 1988; Menning *et al.* 1988; DiVenere & Opdyke 1990, 1991; Lozovsky & Molostovsky 1993; Steiner *et al.* 1993; Opdyke & Channell 1996; Gialanella *et al.* 1997; Nawrocki 1997; Opdyke *et al.* 2000, 2001; Alva-Valdivia *et al.* 2002; Szurlies *et al.* 2003; Steiner 2006; Belica *et al.* 2017).

The start of the KRS has been assigned an age of *ca.* 318 Ma from two U–Pb SHRIMP zircon ages of 317.8 ± 2.8 Ma (Eastons Arm Rhyolite) and 318.0 ± 3.4 Ma (Peri Rhyolite) in the northern Tamworth Belt of eastern Australia (Opdyke *et al.* 2000). The end of the KRS has recently been constrained by a ^{40}Ar–^{39}Ar plateau (plagioclase) age of 265.1 ± 0.5 Ma from the Bumbo Latite near Kiama, Australia (Belica *et al.* 2017). Beginning in the Capitanian (*ca.* 265 Ma), the geomagnetic field transitioned back into a frequently reversing state, and this transition was thought to occur over a short (∼10 Ma) interval (Hounslow 2016). The late Permian through early Triassic, therefore, marks a period of increasing reversal frequency (∼2–4 Ma^{-1}; Hounslow 2016), providing a strong chronostratigraphic marker (Fig. 3).

We present a composite, ∼1500-m-thick magnetostratigraphic section through a large part of the Ecca Group from the Tanqua depocentre, located in the southwestern segment of the Karoo Basin, and provide a test of the published U–Pb isotopic data sets through comparison with the GPTS. Since the geomagnetic field was restricted to a reverse polarity prior to 265 Ma, the presence of reversals would indicate that deposition of the ash beds occurred in the late Permian, whereas a uniform reverse polarity would support the existing biostratigraphic age data, as well as provide strong evidence for disturbance of the U–Pb system.

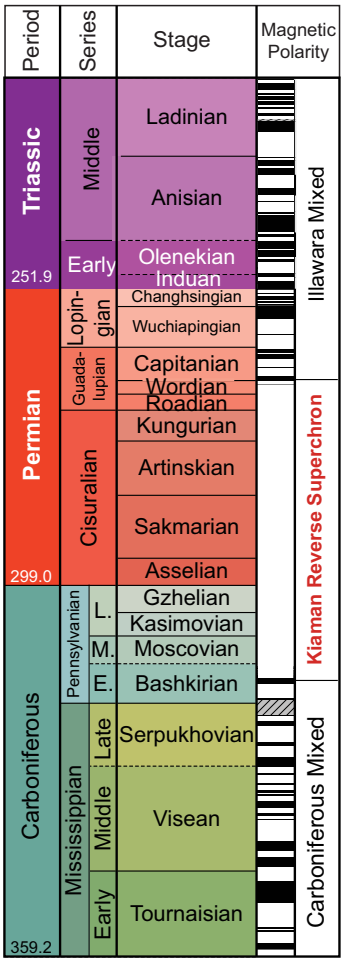


Figure 3. The Geomagnetic Polarity Time Scale (GPTS) modified from Gradstein *et al.* (2012) showing the extent of the Kiaman Reverse Superchron (KRS).

2 GEOLOGICAL BACKGROUND

The Karoo Basin (∼700 000 km^2) has been classically interpreted to be a retro-arc foreland basin that developed behind an inferred magmatic arc and fold-thrust belt (Johnson *et al.* 1996; Catuneanu *et al.* 1998; Milani & De Wit 2008; Lindeque *et al.* 2011). In this interpretation, the basin formed ∼1000 km inland from the southern Panthalassan margin of Gondwana and to the north and east of the two branches of the Cape Fold Belt (Fig. 1). A more recent interpretation by Tankard *et al.* (2009, 2012) suggested that subsidence during the deposition of the Ecca Group and lower Beaufort Group was a function of mantle flow associated with dynamic topography of the subducting plate where foreland basin formation (linked to emergence of the Cape Fold Belt) did not take place until the Triassic. This model is consistent with provenance studies (Van Lente 2004), gross thickness trends, and low rates of deposition in the lower Ecca Group (Flint *et al.* 2011).

Tectonic activity related to the Cape Orogeny has been suggested to have occurred in two separate deformation events, the first at *ca.* 275–260 Ma, and the second at *ca.* 255–245 Ma (Hansma *et al.* 2016). The southern Cape Fold Belt is associated with low-temperature metamorphism and deformation folds that decrease in amplitude northward toward the Karoo foreland basin (Paton *et al.* 2006). Deformation in the Cape Fold Belt has been linked to synchronous sedimentation in the foreland basin (Catuneanu *et al.*

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1998), though the provenance of the Ecca Group has also been interpreted to be the Patagonian batholith (Van Lente 2004). The basin fill succession shows palaeocurrents consistent with a long-term northeasterly progradation of the depositional systems (Johnson *et al.* 1996; Wild *et al.* 2009; Flint *et al.* 2011; Poyatos-Moré *et al.* 2016).

The base of the Karoo Supergroup is defined by the glacigenic Dwyka Group that consists of diamictites formed during the Carboniferous–Permian glaciation of southern Gondwana (Fig. 3). Two U–Pb SHRIMP zircon ages of 288.0 ± 3.0 and 289.6 ± 3.8 Ma reported from tuffs in the overlying Prince Albert Formation (Fig. 2) suggest a *ca.* 290 Ma minimum age for this unit (Bangert *et al.* 1999), and palaeomagnetic data support deposition during the KRS (Opdyke *et al.* 2001). The Lower Ecca Group overlies the Dwyka Group (Fig. 2), and in the Tanqua depocentre (this study location; Fig. 1) is composed of shales to fine grained turbidites of the Prince Albert, Whitehill and Collingham formations, which indicate the transition to prolonged postglacial sea-level rise and the initiation of deep-water deposition (Hodgson *et al.* 2006).

The middle-upper Ecca Group (Fig. 2) comprises ∼1500 m of basin floor to deltaic strata; this includes basinal mudstones of the Tierberg Formation (Fig. 2), four basin floor turbidite fans (Fans 1–4) and a channelized lower slope complex (Unit 5; Fig. 2) of the Skoorsteenberg Formation, overlain conformably by submarine slope/shelf edge strata of the Kookfontein Formation, and deltaic deposits of the Waterford Formation (Wickens 1994; Wild 2005; Hodgson *et al.* 2006; Wild *et al.* 2009; Gomis-Cartesio *et al.* 2016; Poyatos-Moré *et al.* 2016). U–Pb ages published from the Ecca Group yield maximum depositional ages ranging from *ca.* 274–250 Ma (Fildani *et al.* 2007, 2009; McKay *et al.* 2015, 2016). The younger ages in this range (*ca.* 260–250 Ma) are problematic in that they require the PTB to be placed within the middle Ecca Group, as opposed to the biostratigraphically supported placement in the overlying terrestrial fluvio-lacustrine strata of the Beaufort Group (Figs 1 and 2).

Some authors have explained this apparent age inversion by applying a diachronous sedimentation model across the Karoo Basin, where the submarine fans of the Ecca Group in the southwest are considered correlative with the terrestrial Permian–Triassic sections located farther east (Fildani *et al.* 2007, 2009). However, this interpretation is not supported by recent regional mapping that shows continuity of the marine Ecca Group for over 500 km across the basin, with a series of interfingering submarine fan deposits, all overlain by eastward prograding deltaic deposits (Flint *et al.*, personal communication, 2017). This proposed diachronous sedimentation model is also inconsistent with the biostratigraphic constraints on basin infill starting in the southwest during the deposition of the Beaufort Group (Rubidge 2005).

3 SAMPLING AND METHODOLOGY

3.1 Palaeomagnetism

Palaeomagnetic cores were collected from three partially overlapping sections comprising ∼1500 m of the Tierberg, Skoorsteenberg, Kookfontein and Waterford Formations (∼600 samples) of the Ecca Group in the Tanqua depocentre (Figs 1 and 2). The Ecca Group sedimentary rocks sampled here are almost flat-lying with regional dips of 1°–3°, and lie outside the main region of Cape Fold Belt deformation (Figs 1 and 2). Approximately 950 m of the Tierberg and Skoorsteenberg Formations were sampled from the OR1 core with a sampling interval of about 4 m (∼200 samples). Magnetic susceptibility measurements were taken for the top 230 m of section at a resolution of 0.25 m. The overlying Kookfontein (Pienaarsfontein outcrop locality) and Waterford Formations (SL1 outcrop locality) were sampled from two field sections located ∼20 km apart (Fig. 4). Lithologic correlation between the two sections (Fig. 4) was made using the interpreted maximum regional flooding surfaces of Wild *et al.* (2009) and Poyatos-Moré *et al.* (2016). At Pienaarsfontein and SL1, the Kookfontein and Waterford Formations were sampled at a resolution of 2 m (∼300 samples) and magnetic susceptibility measurements were taken with an SM-20 portable magnetic susceptibility meter to determine the variability of magnetic properties along the section at a resolution of 0.25 m.

Samples were drilled in the field using a portable, petrol-powered hand drill and oriented using a Brunton magnetic compass and a solar compass to correct for local magnetic declination. Thickness of the stratigraphic succession was measured using a Jacob's staff perpendicular to bedding to account for tectonic dip, and a detailed stratigraphic log was completed at a 1 m resolution with GPS points tagged every 10 m for correlation with the logged section of Wild (2005). The section was photographed and structural attitudes were measured in order to correct for any post-magnetization tilting. Samples were then cut into 10 cm^3 cylindrical cores, and measured at the Alpine Laboratory of Paleomagnetism in Peveragno, Italy, where a pilot study was conducted to determine the best treatment for isolation of the individual magnetic components. To avoid the acquisition of a viscous remanence over 400 °C, the samples were demagnetized in a field free space and measured directly after cooling.

Stepwise thermal demagnetization was conducted using an ASC TD48 thermal demagnetizer to temperatures up to 670 °C. Samples were measured on a 2 G DC-SQUID magnetometer, and palaeomagnetic directions were isolated via principal component analysis (Kirschvink 1980) using PalaeoMag 3.1 (Jones 2002). Individual components were calculated from three or more points, and directions were restricted to mean angular deviation (MAD) values $\leq 10^\circ$ for lines and $\leq 15^\circ$ for planes in calculation of the mean palaeomagnetic pole. A minimum of three points (four for planes) defined a stable endpoint magnetization, and anchored fits were avoided due to the anomalously low uncertainty estimations (Heslop & Roberts 2016). Where MAD values were greater than 15°, or where samples did not reach stable endpoints, the arc method of McFadden & McElhinny (1988) was used to isolate the primary component of magnetization. Final iterative directions along the arc constraints were combined with lines for the stratigraphic polarity interpretation.

The reversal stratigraphy (declination, inclination, and VGP latitude) was then plotted against sampling level and a composite magnetostratigraphic barcode was completed and compared with the GPTS (Hounslow & Balabanov 2016; Ogg *et al.* 2016). Magnetic reversals were interpreted from remanence directions that crossed the magnetic equator (defined as 90° along a great circle from the mean direction) where three or more consecutive samples define a magnetozone. Single-sample polarity reversals (marked by a half-bar to represent uncertainty) indicate either (1) a geomagnetic subchron; (2) a magnetic reversal with insufficient adjacent sampling, or (3) possible remagnetization of the individual sample. Palaeomagnetic directional data are provided in Table 1.

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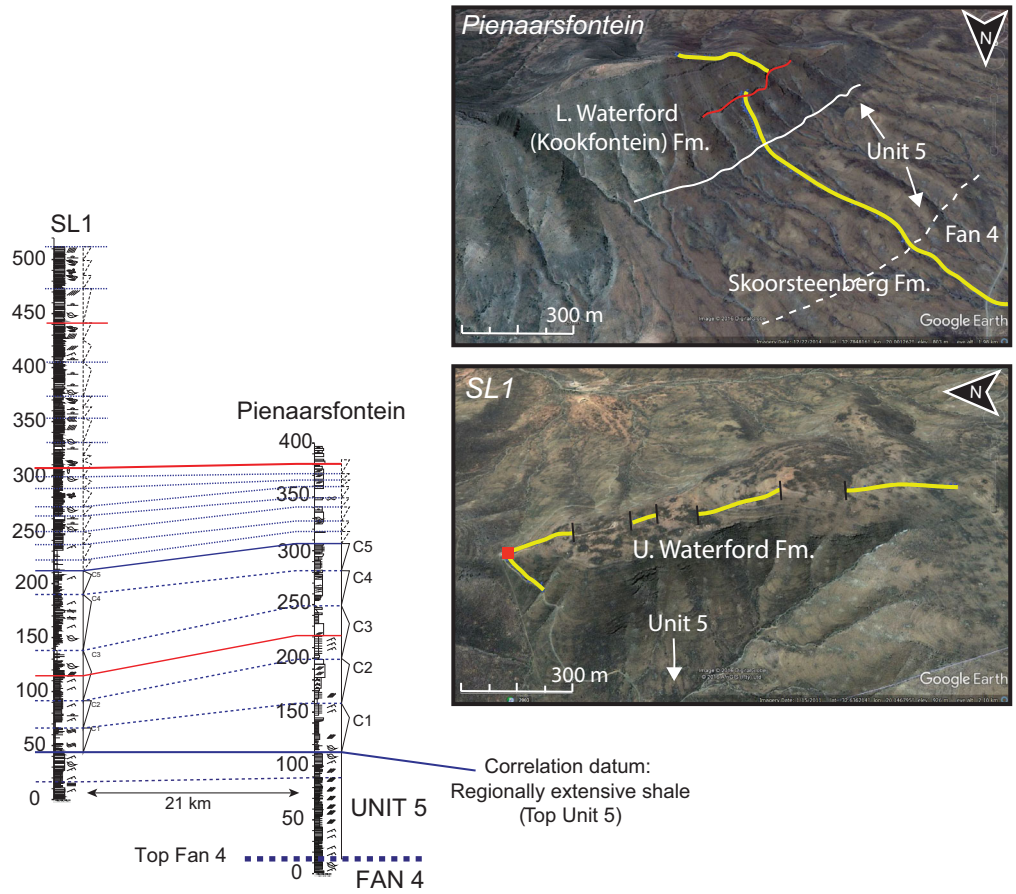


Figure 4. Stratigraphic correlation between the two field sections SL1 and Pienaarsfontein spanning the Kookfontein and Waterford Formation using measured sections from Wild (2005). The top of Unit 5 is represented by a regionally extensive mudstone. Section locations and sampling sites (GPS-tagged) are shown by yellow lines on the Google Earth images. Sections are located approximately 20 km apart.

Table 1. Palaeomagnetic results.

Section	Component	S_{lat} (°S)	S_{long} (°E)	$N(n)$	P	D (°)	I (°)	α_{95}	k	P_{lat} (°N)	P_{long} (°E)	d_p/d_m
SL1	B	32.631	20.145	48	*N*	319.5	−64.5	7.7	8.08	−56.3	74.1	9.9/12.3
(Waterford Fm.)	**C**	**32.631**	**20.145**	**63 (50)**	***R***	**122.9**	**65.5**	**10.0**	**5.05**	**−45.0**	**73.2**	**13.2/16.2**
	**C	32.631	20.145	37	*R*	124.7	65.3	12.9	4.18	−46.2	73.6	16.9/20.9
Pienaarsfontein	A	32.780	19.993	34	*N*	3.1	−69.0	6.47	15.44	−70.2	14.4	9.3/11.0
(Kookfontein Fm.)	B	32.780	19.993	158	*N*	341.0	−60.8	2.50	21.16	−72.4	73.6	2.9/3.8
	C	**32.780**	**19.993**	**163 (149.5)**	***R***	**116.0**	**62.1**	**4.22**	**8.47**	**−39.8**	**78.2**	**5.1/6.6**
	**C	32.780	19.993	136	*R*	114.4	61.8	4.54	8.08	−38.6	78.4	5.4/7.0
Ecca Group (SL1 +Pienaarsfontein)	**C**	**32.706**	**20.069**	226 (199.5)	***R***	**117.3**	**62.8**	**3.98**	**7.27**	**−40.8**	**77.4**	**4.9/6.2**
	**C	32.706	20.069	173	*R*	116.2	62.5	4.46	6.76	−40.0	77.7	5.4/7.0

Notes: S_{lat} = site latitude, S_{lon} = site longitude, N = number of samples used for calculation of the mean direction, $n = L + \frac{1}{2}p$ (L = lines and p = planes; McFadden & McElhinny 1988), P = polarity, D = declination, I = inclination, α_{95} = radius of the cone of 95 per cent confidence about the mean direction, k = precision parameter (Fisher 1953), P_{lat} = pole latitude, P_{long} = pole longitude, d_p/d_m= semi-minor and semi-major axes of oval of 95 per cent confidence, ** = mean excluding arcs and bold = recommended palaeomagnetic mean.

3.2 Magnetic fabric

Eighty-seven samples were processed from the Ecca Group (Pienaarsfontein and SL1) to determine the anisotropy of magnetic susceptibility (AMS). Principal axes of AMS were measured on a MFK1-FA Kappabridge using the three-axis spinning protocol at the University of Western Australia. Results were plotted using the Anisoft 42 software (Chadima & Jelínek 2009), and analysed using Jelínek statistics (Jelínek & Kropáček 1978; Jelínek 1981). Results are provided in Table 2.

3.3 SHRIMP U–Pb geochronology

Ten ash beds from the Tierberg and Skoorsteenberg Formations of the Ecca Group were sampled and processed for geochronological work. Samples were collected from the continuous 800-m-thick core of the research borehole OR1 (Fig. 1). Ash layers are typically 2–6 cm thick, and consist of fine-grained siliciclastic and well-consolidated clay-rich volcaniclastic material interbedded within siltstones. Zircon grains were isolated from each ash layer using conventional mineral separation techniques at the University of

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Table 2. AMS results.

Site	N	K_m	L	F	P	T	U	$K_{max}D$	$K_{max}I$	$K_{int}D$	$K_{int}I$	$K_{min}D$	$K_{min}I$
Pienaarsfontein	60	1.97e−4	1.005	1.027	1.033	0.669	0.665	126.1	19.0	216.8	2.0	312.6	70.9
SL1	27	1.50e−4	1.008	1.015	1.023	0.310	0.306	93.0	7.8	259.0	82.0	2.7	1.9
Ecca mean*	87	1.83e−4	1.006	1.023	1.030	0.558	0.553	107.8	17.3	202.2	13.8	328.8	67.6

Notes: N = number of samples; K_m = average bulk volume susceptibility in SI units; L = magnetic lineation: K_{max}/K_{int}; F = foliation: K_{int}/K_{min}; P = anisotropy degree: K_{max}/K_{int}; T = shape factor: $T = 2\ln(K_{int}/K_{min})/\ln(K_{max}/K_{min}) - 1$; U = difference shape factor: $(2K_{int} - K_{max} - K_{int})/(K_{max} - K_{int})$; K_{max} (K_{min}); Dec/Inc = Declination/Inclination.

*Mean tensor directions were calculated from combined sample-level data. (Jelínek & Kropáček 1978; Jelínek 1981).

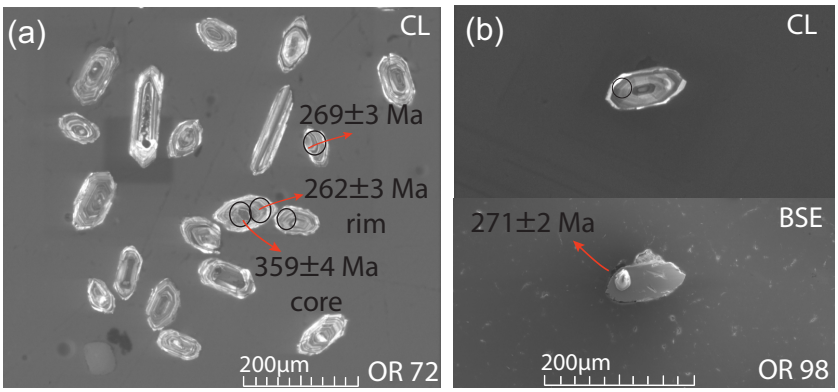


Figure 5. Backscattered electron (BSE) and cathodoluminescence (CL) images of zircon grains from the lower Ecca Group (OR1 core). SHRIMP spots are shown by the black ovals. (a) OR 72 (331.1 m) and (b) OR 98 (452.8 m).

Western Australia. Each lithified ash sample was pulverized with a rock hammer, processed through a disc mill, washed and placed in a sonic bath for 30 min and dried. The crushed material was separated using 40 and 150 µm sieves and a bar magnet to remove ferromagnetic components before processing through a Frantz Isodynamic Separator. The non-magnetic portion was then put through a heavy liquid density separation with multiple agitation periods using LST (Lithium heteropolytungstates) heavy liquid (ρ = 2.85 g cm^{-3} at 25 °C) for 4 hr to isolate grains in the higher density fractions. The heavy fraction was washed and handpicked under a binocular microscope.

Six of the 10 ash beds yielded zircon grains considered suitable for analysis. Within the six samples, three main populations of zircon morphology were identified, including a population of idiomorphic and elongate grains with oscillatory cathodoluminescent zoning and an aspect ratio of 4:1 (Fig. 5a). Two other populations of euhedral zircons with oscillatory zoning were also sampled with aspect rations of 2–3:1 (Figs 5a and b). Around 50 randomly picked zircon grains from each sample were mounted alongside the zircon standards M257 (561.3 Ma; 840 ppm U; Nasdala *et al.* 2008) and Temora 2 (416.8 Ma; Black *et al.* 2004) in 2.5 cm diameter epoxy discs (3–4 samples per disc) as external standards for U/Pb ratios and U concentrations. Discs were then polished and gold coated (20 µm) for imaging with the backscatter electron (BSE) and cathodoluminescence (CL) detectors of a Tescan Vega 3 analytical SEM system at the Centre for Microscopy, Characterization and Analysis at the University of Western Australia. Microphotographs were used to identify magmatic features such as oscillatory zoning and euhedral/elongate form (Fig. 5). Apparent inherited cores were identified from CL images as well as fractures and damaged grains (Fig. 5). Highly fractured and metamict grains can produce unreliable U–Pb age data, and were avoided in the subsequent SHRIMP analysis.

The discs were then cleaned using ethanol, petroleum spirits and a soap solution, and rinsed with deionized water. Discs were dried in an oven (1 h at 60 °C) and the surface was gold coated at 40 µm and processed on the SHRIMP II A at Curtin University, Western Australia, following the procedure described by Wingate & Kirkland (2017). Analytical conditions and operational procedures have previously been described (Compston *et al.* 1984; Kennedy & De Laeter 1994; Claoué-Long *et al.* 1995; De Laeter & Kennedy 1998; Wingate & Kirkland 2017). Rastering of the O^{2-} beam was conducted for 120–150 s to reduce common Pb contamination before each analysis, and each spot was focused using a primary ion beam of 1.5–1.9 nA with a diameter of ∼25 µm. SHRIMP spots were made on both inherited cores and rims to determine the level and types of inheritance, and a common Pb correction was applied using the measured amount of ^{204}Pb. Data reduction and correction were conducted as described by Wingate & Kirkland (2017) using the SQUID2 and Isoplot3 software (Ludwig 2008, 2009). All calculated ages are reported at 95 per cent confidence. To reduce the problems caused by zircon grains with potentially disturbed U–Pb systems, analyses that comprised more than 1.5 per cent of non-radiogenic ^{206}Pb or U concentrations over 1000 ppm were rejected. Analytical data are presented in Table 3.

4 RESULTS AND INTERPRETATION

4.1 Palaeomagnetism

4.1.1 Ecca Group, Pienaarsfontein

The outcrop section spanning the upper Skoorsteenberg and Kookfontein Formations (Fig. 4) displayed the least complicated thermal demagnetization spectra of all the studied sections

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Table 3. U–Pb SHRIMP results.

Spot	$^{206}Pb_c$ (per cent)	U (ppm)	Th (ppm)	$^{232}Th/^{238}U$	±%	$^{207}Pb/^{206}Pb$	±%	$^{206}Pb/^{238}U$	±%	$^{206}Pb/^{238}U$ (Age) (1)
Sample OR 98										
D-8	0.32	613	787	1.33	0.16	0.0546	1.3	0.0590	3.6	265 ± 2
D-7	0.31	223	118	0.55	0.31	0.0540	2.1	0.0652	1.3	266 ± 2
D-3	0.07	867	675	0.80	0.62	0.0533	1.0	0.0775	2.9	266 ± 2
D-10	0.27	497	278	0.58	0.49	0.0585	1.2	0.0681	2.9	268 ± 3
D-4	0.03	814	972	1.23	0.28	0.0534	1.0	0.0806	2.0	268 ± 2
D-5	0.55	425	312	0.76	0.30	0.0581	1.4	0.0738	2.5	268 ± 3
D-13	1.21	636	696	1.13	0.45	0.0654	1.1	0.0676	5.0	269 ± 3
D-6	0.93	377	330	0.90	0.74	0.0623	1.6	0.0558	6.4	270 ± 2
D-14	0.07	401	254	0.65	0.22	0.0539	2.4	0.0770	3.4	270 ± 2
D-11	0.74	574	579	1.04	0.16	0.0530	1.3	0.0727	3.0	271 ± 4
D-15	0.28	892	667	0.77	0.27	0.0565	1.0	0.0554	4.2	271 ± 2
D-1	0.29	518	383	0.76	0.81	0.0550	1.3	0.0717	3.2	271 ± 2
D-12	0.45	668	802	1.24	0.43	0.0534	1.2	0.0706	1.7	272 ± 2
D-9	0.60	449	510	1.17	1.12	0.0537	2.3	0.0739	2.8	272 ± 3
Rejected analyses										
D-16	0.25	909	928	1.05	0.36	0.0543	1.0	0.0657	6.2	260 ± 4
D-2	0.40	381	292	0.79	0.64	0.0560	1.5	0.0704	3.3	261 ± 2
D-17	1.84	370	418	1.17	1.20	0.0730	1.3	0.0754	2.7	264 ± 2
D-1	1.98	194	111	0.59	0.65	0.0678	3.8	0.0713	1.2	267 ± 3
Sample OR 72										
OR72–18	0.55	359	176	0.51	0.26	0.0592	2.0	0.0636	1.3	258 ± 3
OR72–9	0.00	451	248	0.57	0.67	0.0527	1.9	0.0635	0.8	259 ± 3
OR72–13	0.07	500	271	0.56	0.22	0.0532	1.8	0.0713	1.9	260 ± 5
OR72–7	0.08	430	211	0.51	0.23	0.0536	1.8	0.0700	1.8	262 ± 3
OR72–17	0.27	438	177	0.42	0.25	0.0531	1.9	0.0629	2.0	263 ± 3
OR72–2	0.09	403	229	0.59	0.70	0.0543	1.9	0.0667	1.7	263 ± 4
OR72–5	0.31	612	408	0.69	1.18	0.0519	1.5	0.0721	1.9	263 ± 4
OR72–6	0.06	627	407	0.67	0.28	0.0504	2.7	0.0671	1.1	263 ± 3
OR72–1	0.14	695	530	0.79	0.16	0.0510	1.5	0.0709	0.9	265 ± 4
OR72–3	0.28	384	529	1.42	0.92	0.0520	1.7	0.0796	0.7	265 ± 4
OR72–15	0.21	484	253	0.54	0.62	0.0513	1.7	0.0723	1.2	266 ± 4
OR72–4	0.00	414	265	0.66	0.53	0.0538	1.7	0.0768	1.5	269 ± 3
OR72–14	0.23	458	188	0.42	0.24	0.0514	1.8	0.0723	2.0	270 ± 4
OR72–11	0.19	380	183	0.50	0.68	0.0531	2.0	0.0678	0.8	276 ± 3
Inherited population										
OR72–8	0.08	1394	361	0.27	0.65	0.0635	0.8	0.0858	3.0	359 ± 4
OR72–10	0.03	576	272	0.49	0.19	0.0709	0.8	0.2330	1.6	768 ± 9
Rejected analyses										
OR72–12	0.23	322	149	0.48	0.26	0.0530	2.2	0.0587	0.9	248 ± 3
OR72–16	0.00	1070	756	0.73	0.24	0.0532	2.3	0.0714	2.0	264 ± 3
Sample OR 105										
OR-15	0.91	134	102	0.79	0.81	0.0566	3.4	0.0632	1.4	251 ± 4
OR-7	1.36	139	145	1.08	0.38	0.0667	3.0	0.0617	1.3	252 ± 4
OR-12	0.31	362	471	1.34	1.14	0.0561	2.0	0.0679	2.2	253 ± 3
OR-10	0.96	266	217	0.85	0.26	0.0595	2.3	0.0626	3.1	257 ± 3
OR-8	1.26	276	273	1.02	0.42	0.0582	2.3	0.0654	1.0	258 ± 4
OR-1	0.52	198	204	1.06	0.65	0.0571	2.6	0.0775	1.1	259 ± 6
OR-3	0.63	392	450	1.19	0.21	0.0549	2.1	0.0621	0.8	260 ± 4
OR-2	0.58	294	244	0.86	0.24	0.0543	2.2	0.0697	2.1	261 ± 4
OR-4	0.68	220	393	1.84	0.56	0.0588	3.8	0.0729	1.0	272 ± 5
OR-11	3.14	322	311	1.00	1.12	0.0762	4.4	0.0413	4.4	296 ± 5
OR-9	0.18	1399	365	0.27	1.11	0.0603	0.7	0.1271	1.1	425 ± 6
OR-13	0.08	501	548	1.13	0.35	0.0571	1.3	0.1254	1.2	445 ± 5
OR-14	0.16	680	107	0.16	0.48	0.0577	1.1	0.1175	2.6	450 ± 6
0R-6	0.64	290	324	1.16	0.38	0.0545	2.3	0.0726	2.2	259 ± 3
0R-5	0.53	420	309	0.76	0.23	0.0617	1.5	0.1006	0.7	444 ± 5
Sample OR 107										
C-7	0.60	1764	614	0.36	0.96	0.0595	0.7	0.0596	3.9	224 ± 2
C-9	0.19	1793	819	0.47	0.11	0.0556	0.7	0.0689	3.3	251 ± 2
C-4	0.08	365	64	0.18	0.33	0.0514	1.6	0.0762	1.7	271 ± 2
C-3	0.42	1393	78	0.06	0.93	0.0638	0.6	0.1139	6.2	380 ± 2
C-11	0.00	1059	558	0.54	0.35	0.0548	0.7	0.1209	1.8	389 ± 2

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Table 3. (*Continued.*)

Spot	$^{206}Pb_c$ (per cent)	U (ppm)	Th (ppm)	$^{232}Th/^{238}U$	±%	$^{207}Pb/^{206}Pb$	±%	$^{206}Pb/^{238}U$	±%	$^{206}Pb/^{238}U$ (Age) (1)
C-2	0.15	109	45	0.42	0.45	0.0579	2.0	0.1301	3.2	468 ± 5
C-6	0.37	298	83	0.29	0.31	0.0566	2.2	0.1403	2.6	470 ± 4
C-5	0.04	1012	141	0.14	1.16	0.0631	0.7	0.1104	4.3	478 ± 4
C-1	0.14	459	156	0.35	11.60	0.0617	0.8	0.1920	1.4	589 ± 11
C-8	0.11	527	206	0.40	0.21	0.0758	0.6	0.2570	2.9	884 ± 6
C-10	0.23	104	132	1.31	0.56	0.0802	1.1	0.3349	3.1	1136 ± 13

Notes: Errors are 1σ; Pb_c indicates the common Pb portions. Error in standard calibration was 0.40 per cent (not included in above errors but required when comparing data from different mounts) for OR 72 and OR 105 and 0.22 per cent for OR 98 and OR 107. (1) Common Pb corrected using measured ^{204}Pb. Bold = Samples with calculated ages (Fig. 13).

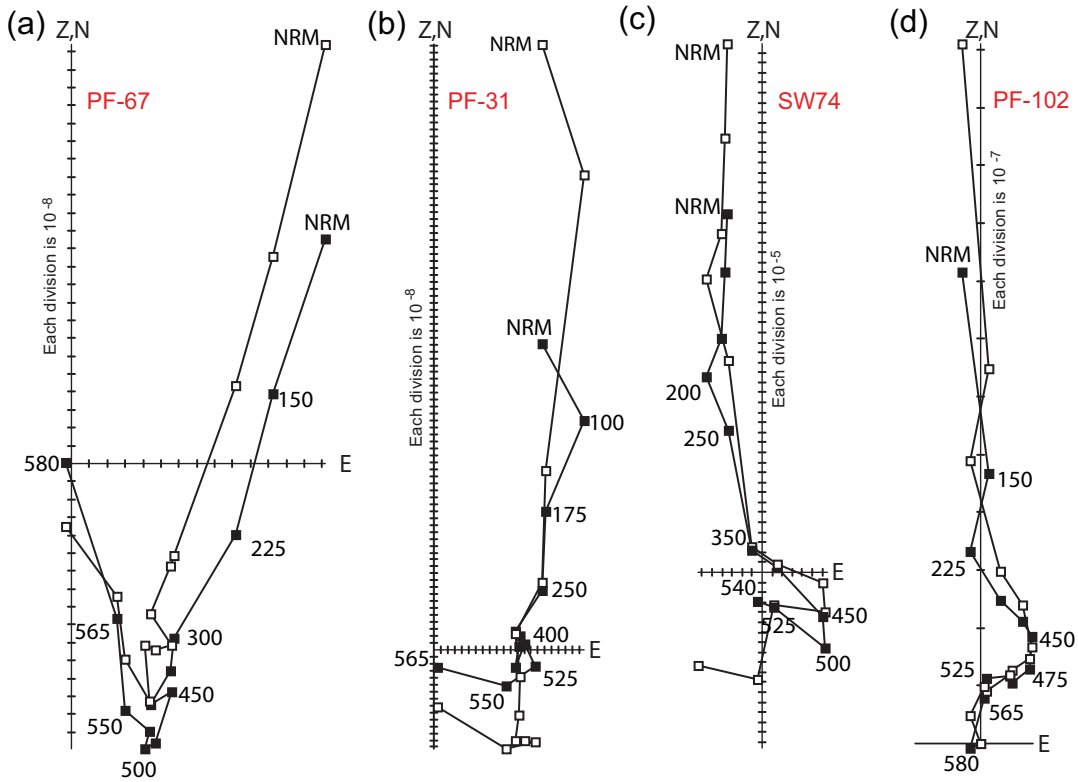


Figure 6. Representative Zijderveld demagnetization diagrams for the Waterford Formation. (a) and (b) Reverse polarity samples from the Kookfontein Fm. (Lower Waterford) at Pienaarsfontein, with removal of the Jurassic overprint direction at 475°C; (c) reverse polarity sample from the Upper Waterford Fm. at SL1 showing the removal of two normal polarity overprint directions before isolation of the ChRM from 525–540 °C and (d) normal polarity sample from the upper Kookfontein Fm. at Pienaarsfontein showing the removal of two overprint components before isolation of the ChRM at 580 °C. Solid (open) squares represent projections on the horizontal (vertical) plane. For the accompanying, equal area projections and intensity decay diagrams see Figs 7 and 8.

(Figs 6–8a, b and d). Mean palaeomagnetic directions are reported in Table 1. The natural remanent magnetization (NRM; magnetization before demagnetization) intensities of the sedimentary rocks range from 0.3 to 113 mA m^{-1}, with an average value of 4.0 mA m^{-1}. NRM intensities were highest for the top 50 m of section (∼1430 mA m^{-1}), and randomly oriented and shallow directional data suggest that this segment has been completely remagnetized by lightning. These uppermost samples were discarded from further consideration. Magnetic susceptibility data revealed a mainly ∼300 m rhythmic susceptibility profile with an average value of 0.15 (× 10^{-3} SI units; Fig. 9).

Progressive thermal demagnetization of the samples revealed up to three discernible palaeomagnetic components, with all samples becoming unstable at $T > 580$ °C (Figs 6–8). The first, referred to as Component A, is present in about 20 per cent of the samples, and is stable from about 110–250 °C. The mean direction is broadly consistent with a present field overprint, although we note that the small number of isolated directions may not be representative of the true mean (Fig. 10a). In some cases, this component is absent or is preceded by a randomly oriented low-temperature direction stable below 150 °C that may be representative of goethite or a viscous remanent magnetization (VRM). The second component, referred to as Component B, is present in nearly every sample, and is stable from about 300–450 °C (Figs 7–9). Directional data (declination = 341.0°; inclination = −60.8° and α_{95} = 2.5) are similar to both reported Jurassic overprint directions and a modern field component (De Kock & Kirschvink 2004; Lanci *et al.* 2013; Moulin *et al.* 2017). When both Components A and B are recorded by the same sample, Component B always shows higher unblocking temperatures, and is the last direction isolated before the unblocking of the characteristic remanent magnetization (ChRM) direction (Component C), which is interpreted to be primary (Fig. 10b). Component C (D = 116.0°; I = 62.1° and α_{95} = 4.2°) is stable from about 475–580 °C, and is restricted to reverse polarity, with the

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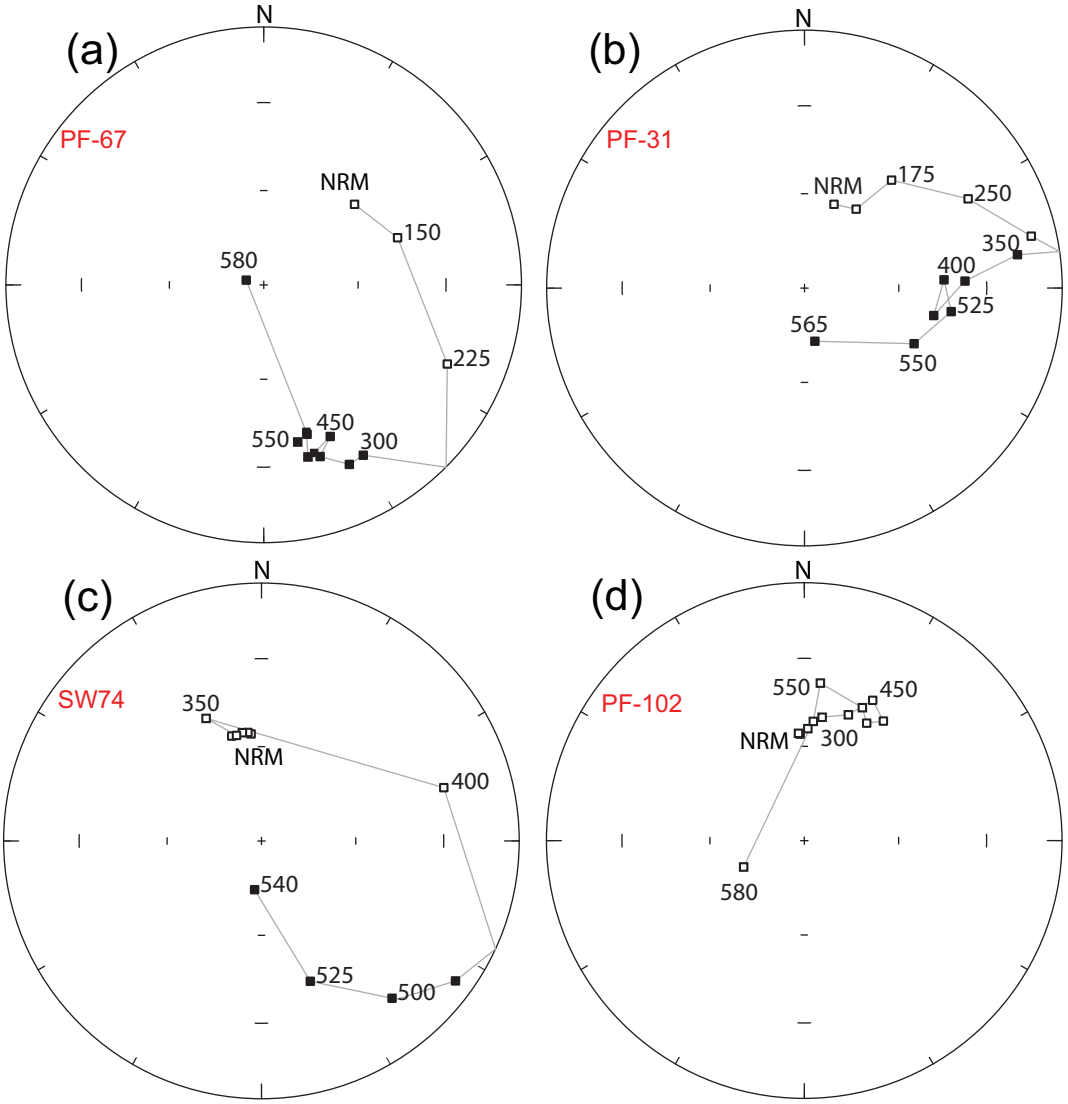


Figure 7. Representative equal area projections for the Waterford Formation. Up (down) pointing palaeomagnetic directions are indicated by open (closed) squares. Numbers represent the thermal demagnetization steps in °C.

exception of a few samples throughout the section which display normal polarity. Some of these normal polarity samples showed stable endpoint magnetizations isolated from 565–580 °C after the removal of both normal Components A and B (Fig. 7d), and may be correlative with previously documented *ca.* 275 Ma normal subchrons within the KRS (Hounslow & Balabanov 2016).

Intensity decay diagrams show demagnetization curves consistent with magnetite as the main magnetic phase (Fig. 8). Isolation of this ChRM component within the unblocking temperatures of magnetite excludes secondary haematite as a possible source of magnetization (Butler 1992). Components B and C are statistically distinct and fail a common distribution test at the 95 per cent confidence level ($\lambda_o = 21.1°$, $\lambda_c = 7.0°$; McFadden & McElhinny 1990). Previous palaeomagnetic work in the Tanqua depocentre revealed up to three palaeomagnetic components, including a low-temperature ($\leq$175 °C) recently acquired VRM and a higher temperature ($\leq$475 °C) remagnetization that may be related to Jurassic large igneous province (LIP) emplacement (Lanci *et al.* 2013; Tohver *et al.* 2015). Reverse polarity ChRM data reported from the overlying Abrakamskraal Formation in the Beaufort Group (Fig. 2) were similarly isolated at temperatures of 450–580 °C (Lanci *et al.* 2013).

The mean equal area projection of the ChRM directions (Fig. 10b) shows a somewhat N-S elongation. No pattern in stratigraphic level was observed with respect to this elongation, or any difference in directional data when Bingham statistics were applied. Typically inclination shallowing expected from sedimentary rocks which have undergone compaction is recognized by an E-W directed elongation of the palaeomagnetic data set (Tauxe & Kent 2004). This section lies within a geomagnetic superchron, and the exact behaviour of the Earth's magnetic field during these types of events is still poorly understood (Granot *et al.* 2012; Hounslow 2016; Lhuillier *et al.* 2016). Contributions from a non-zero non-GAD (Geocentric Axial Dipole) field can change the distribution of directions, where a non-zero axial octupole of the same sign as the axial dipole will enhance the N-S elongation of the observed directional data resulting in a shallow polarity bias (Tauxe & Kent 2004). The superchron recorded here was of reverse polarity, and it has been documented that at least for the past 5 Myr, the reverse polarity field has displayed a larger non-dipole component than the normal field (Opdyke & Henry 1969; Merrill & McElhinny 1977; Schneider & Kent 1988; Johnson *et al.* 2008), which

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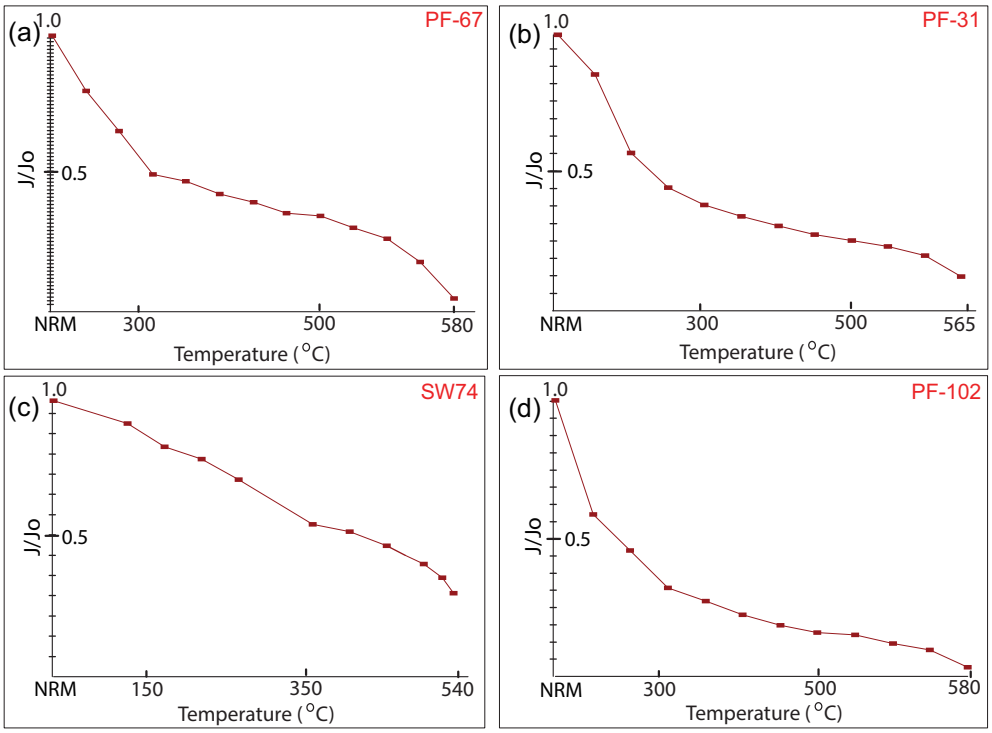


Figure 8. Representative intensity decay diagrams for the Waterford Formation. Demagnetization curves are consistent with magnetite as the main magnetic phase.

may be attributable to a persistent axial octupole contribution (Parés & Van der Voo 2013), though we note this observation is speculative.

4.1.2 Ecca Group, SL1

The outcrop section spanning the upper Waterford Formation at the SL1 locality reveals more complicated demagnetization spectra (Figs 6–8c). The average NRM intensity of this section is slightly higher than that recorded from the Pienaarsfontein section, with an average value of 40 mA m^{-1}.The magnetostratigraphic profile was completed through the stratigraphic succession with some offsets due to lack of exposure along the ridgeline leading up to the SL1 research well (Fig. 4). The third segment of this sampling transect (425–475 m) seems to have been mostly remagnetized by lightning, with elevated NRM intensities and randomly oriented and shallow directional data. Magnetic susceptibility data collected toward the base of the section shows a less regular pattern than that of the Kookfontein Formation, and ranges from 0.03 to a peak of 0.35 ($\times 10^{-3}$ SI units) at the 375 m mark (Fig. 9).

Progressive thermal demagnetization of the samples revealed two distinct palaeomagnetic components identical to Components B and C from the Kookfontein Formation (Pienaarsfontein locality). A few samples exhibit a random low-temperature direction up to 150 °C, however, this overprint is harder to detect here due to the generally less coherent demagnetization data. Component B is present at temperatures from 150 to 350 °C, and is removed above 400 °C (Figs 6–8c). Component C is stable from 475 to 575 °C, and is fully demagnetized above 580 °C (Figs 6–8c). The demagnetization curves from this section indicate some combination of magnetite and maghemite as the main magnetic phases (Fig. 8c). An equal area projection of the mean ChRM data shows a Fisherian distribution, with the reverse polarity sites (positive inclination) displaying a higher scatter in direction due to the overlap in unblocking spectra with the normal polarity overprint (Figs 11a and b). This section is also restricted to reverse polarity, with two potential normal subchrons from two single samples (Fig. 9).

4.1.3 Ecca Group, OR1 core

The samples collected from the long core of the OR1 research well (Fig. 1) show similar demagnetization behaviour to Pienaarsfontein and SL1, with the exception that these samples generally became unstable at temperatures greater than 500 °C. This is attributed to partially fractured samples that were destroyed during heating above 500 °C (before they were completely demagnetized). However, the majority of samples had recoverable remanences and showed a clear removal of a normal polarity overprint direction (Component B) below 475 °C, with isolation of a reverse polarity ChRM direction (Component C) at temperatures ranging from 500 to 580 °C (Fig. 12). The drill core is vertically oriented, so only inclination data are available from this section. Negative inclinations represent normal polarity and positive inclinations represent reverse polarity. This section is restricted to reverse polarity, with one normal polarity remanence (stable at high temperatures) at 200 m depth (Fig. 13). The top of the OR1 core contains the upper sandstones of Fan 4 as well as Unit 5 from the Skoorsteenberg Formation which overlaps with the first 115 m of section at Pienaarsfontein (Wild *et al.* 2009; Poyatos-Moré *et al.* 2016). Both overlapping sections show good agreement with respect to the variation in inclination values, with the steepest inclinations observed near the top of Unit 5, a steady and shallower average inclination of ∼50° for the base, and an average inclination value of ∼60° for the uppermost part of Fan 4 (Figs 9 and 13). Magnetic susceptibility measurements range from −0.2 to 0.4 ($\times 10^{-3}$ SI units), with the largest recorded values correlative with mudstone intervals against

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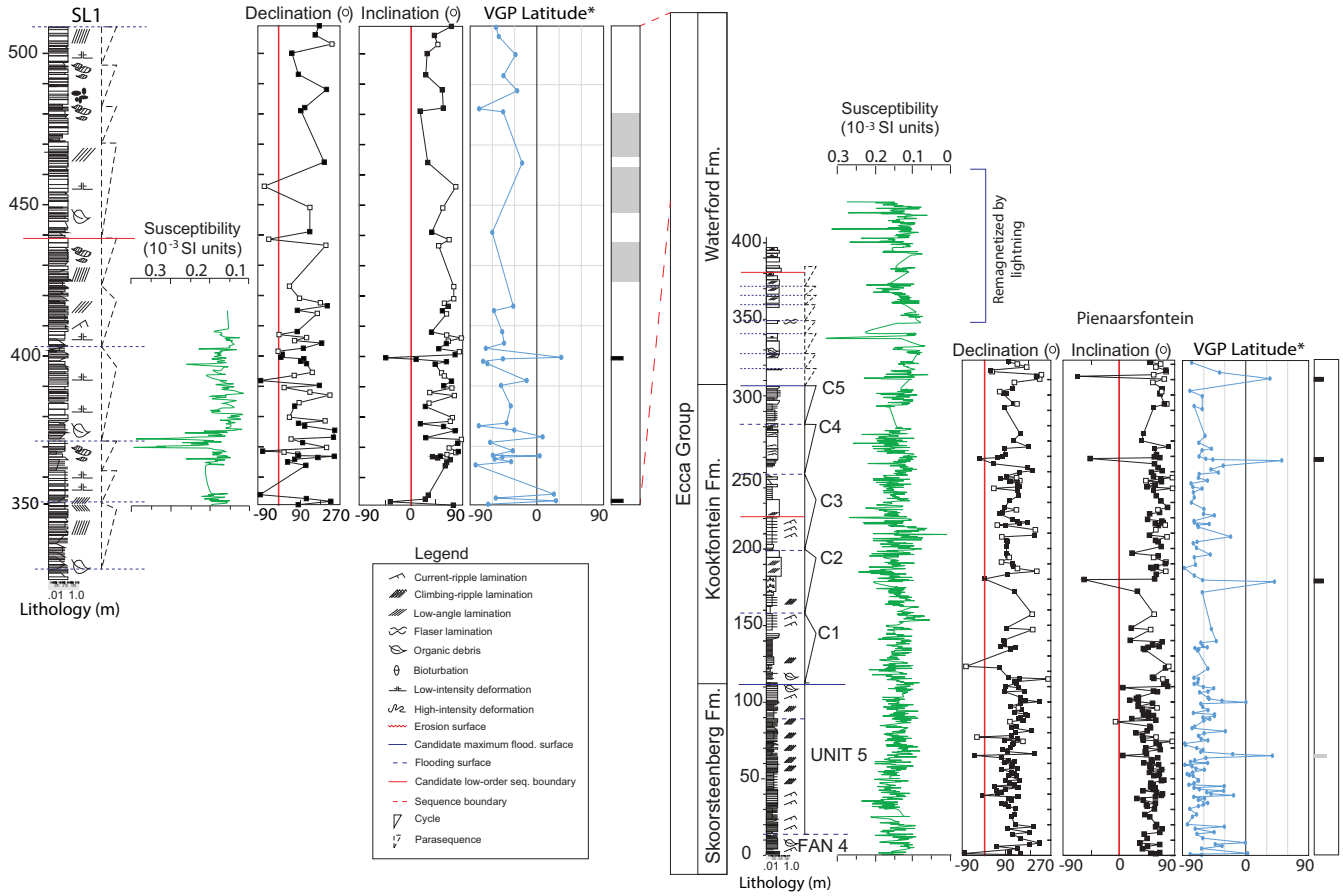


Figure 9. Magnetostratigraphy of the upper Ecca Group (Kookfontein and Waterford Formations) from the field sections SL1 and Pienaarsfontein. ChRM declination and inclination (polarity) data are plotted against sampling level, VGP latitude, lithology and magnetic susceptibility measurements (SI units). The section is restricted to reverse polarity (white), with three potential normal subchrons (black). Both sections show intervals affected by lightning strikes which are shown as grey bars. See Fig. 4 for sequence correlation. Measured sections are from Wild (2005). Closed (open) squares represent directions isolated using lines (arcs).

a rhythmic background susceptibility profile centred at 0.15 ($\times 10^{-3}$ SI units) similar to the pattern observed at Pienaarsfontein (Fig. 9). Overlapping segments of magnetic susceptibility measurements are in strong agreement for Unit 5 (Figs 9 and 13). NRM intensity of the drill-core ranges from 0.25 to 240 mA m^{-1}, with an average value of 3.5 mA m^{-1}.

4.2 Magnetic fabric

AMS results for the Ecca Group are displayed in Fig. 14 along with the corresponding Jelínek statistics and mean directions in Table 2. At Pienaarsfontein (Kookfontein Fm.), the measured anisotropy factors do not show any significant variation along section or for any particular lithology. The shape of the AMS ellipsoid is oblate, with a shape factor (T) = 0.669 (Fig. 14a). The fabric shows K_3 values clustered around the vertical axis consistent with horizontally layered sediments unaffected by ductile deformation (Fig. 14a). The similarity between the shape factors (T) and difference shape factors (U) indicates a very low degree of anisotropy for the sedimentary rocks (Jelínek 1981). At SL1 (Waterford Fm.), the AMS ellipsoid shows a mostly oblate shape (Fig. 14b); however, the Jelinek diagrams indicate a more complex geometry than expected from typical sedimentary burial and compaction (T = 0.310). The K_3 values here are much shallower (<35°), and are clustered in an N–S direction (Fig. 14b). This fabric may be a result of an intersection lineation between subhorizontal bedding and a vertical foliation (Parés *et al.* 1999).

4.3 SHRIMP U–Pb geochronology

Of the four samples analysed by SHRIMP (Table 3), three samples yielded at least 10 zircon grains with ages <300 Ma. Two of these samples yielded internally consistent mean weighted $^{206}Pb/^{238}U$ ages suitable for interpretation and are discussed below.

For sample OR72 (331.1 m), 14 standard spots (two rejected from initial instrument calibration) were taken on M257 (MSWD = 3.19), with an external spot-to-spot error of 0.95 per cent(1σ) and a $^{238}U/^{206}Pb$ calibration uncertainty of 0.5178 per cent (1σ). Of 18 analyses from zircon grains, two spots of *ca.* 360 and 770 Ma were interpreted as inherited grains (Table 3). $^{232}Th/^{238}U$ ratios did not indicate any metamorphic resetting, with a minimum observed value of 0.42. One grain was rejected due to a high concentration of U (1070 ppm; Table 3). No grains were rejected based on the proportion of ^{206}Pb; however, one grain (*ca.* 248 Ma) was rejected based on presumed Pb-loss, likely resulting from a nearby fracture in the grain. The remaining 14 spots gave a weighted-mean $^{206}Pb/^{238}U$ age of 264.6 ± 2.9 Ma with an MSWD of 2.25. This age is interpreted here as a minimum constraint due to suspected Pb-loss. A linear trend from the oldest to the youngest grains is observed subparallel to Concordia, and the range in zircon ages

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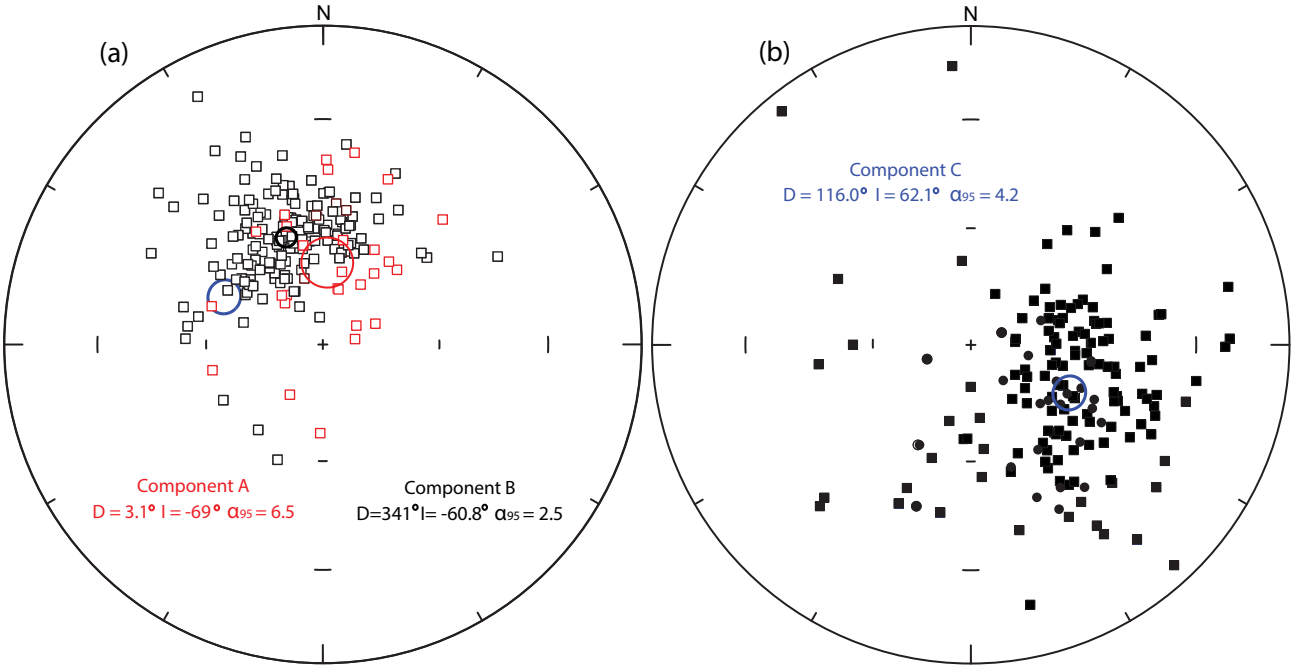


Figure 10. Equal area diagrams displaying the (a) mean overprint directions (Components A and B) and the (b) reverse ChRM direction (Component C) from the Lower Waterford (Kookfontein) Formation at Pienaarsfontein. Up (down) pointing palaeomagnetic directions are indicated by open (closed) squares in the stereoplots. Arc constraints for samples that do not reach stable endpoints were calculated with the method of McFadden & McElhinny (1988), and are shown with circle symbols. Red, blue, and black ovals represent the cone of 95 per cent confidence about the mean direction.

prevents the identification of a single and discrete age population (Fig. 13).

For sample OR98 (452.8 m), nine standard spots (two rejected from initial instrument calibration) were taken (MSWD = 1.10), with an external spot-to-spot error of 0.50 per cent (1σ) and $^{238}U/^{206}Pb$ calibration uncertainty of 0.4973 per cent (1σ). $^{232}Th/^{238}U$ ratios did not indicate any metamorphic resetting, with a minimum observed value of 0.55. Of 18 unknown zircon analyses, two were rejected based on CL images that showed the beam diameter overlapped with fractures. Two analyses were rejected based on ^{206}Pb values (>1.5 per cent; Table 3). No analyses were rejected based on U concentration or inheritance. The remaining 14 spots revealed a distinct age population at *ca.* 271 Ma, with a subpopulation at *ca.* 266 Ma that is likely a result of Pb-loss (Fig. 13). The 14 spots yielded a weighted-mean $^{206}Pb/^{238}U$ age of 269.5 ± 1.2 Ma with an MSWD of 1.02. This calculated age is in good agreement with both the reported magnetostratigraphy (Figs 9 and 13) and the GPTS (Fig. 3), and so provides a good age estimate for the base of the Skoorsteenberg Formation in the Tanqua depocentre.

5 DISCUSSION

5.1 Magnetostratigraphy

Individual magnetostratigraphic columns for each section are plotted in Figs 9 and 13, with a composite of the Ecca Group and overlying Abrahamskraal Formation (Beaufort Group) presented in Fig. 15. The magnetostratigraphy shows an extended reverse polarity interval for over 1500 m of section that is consistent with formation during the Pennsylvanian–Middle Permian KRS (Irving & Parry 1963; Menning *et al.* 1988; Opdyke *et al.* 2000; Steiner 2006; Hounslow & Balabanov 2016). This interval of constant magnetic polarity was followed by a return to a frequently reversing field at *ca.* 265 Ma (Belica *et al.* 2017), thus providing a minimum age constraint for the Ecca Group (Fig. 15).

To provide the best age estimate for deposition of the Ecca Group in the Tanqua depocentre (20°E), we compare this section with the GPTS, as well as the published isotopic ages that do not conflict with the resolved magnetostratigraphy (i.e. >265 Ma; Fig. 15). The normal polarities reported here cannot be confirmed as primary as they are only represented by one consecutive sample; however, they are broadly consistent with the occurrence of normal polarity subchrons within the KRS (Fig. 15).

Our composite 1500 m section from the Ecca Group terminates within the upper Waterford Formation, which underlies a previously published section at Ouberg Pass (upper Waterford-lower Abrahamskraal Formation; Fig. 15) that shows mostly stable reverse polarity endpoints (80 per cent) in the magnetostratigraphy similar to Pienaarsfontein (Lanci *et al.* 2013). Normal polarity magnetozones were also reported (2–3 consecutive samples) for this section, and passed a reversal test. U–Pb SHRIMP (zircon) ages range from *ca.* 262–268 Ma, and the section was interpreted to span ~7 Ma (Lanci *et al.* 2013). Hounslow & Balabanov (2016) proposed two magnetostratigraphic options for this section. The first is that the reported normal polarities from Ouberg Pass are correlative with a late Wordian magnetostratigraphy, with a documented normal subchron at *ca.* 269 Ma. The second is that the normal magnetozones are correlative with post-Kiaman reversals at *ca.* 265 Ma.

U–Pb SHRIMP (zircon) ages of *ca.* 290 Ma are reported from the Prince Albert Formation (lowermost Ecca Group; Fig. 15), while ages of *ca.* 262–268 Ma are reported from the lower Beaufort Group (above the Ecca Group; Fig. 15). A U–Pb SHRIMP (zircon) age of 270.4 ± 2.7 Ma has been reported (Fig. 15) from the middle Tierberg Formation (McKay *et al.* 2016), as well as an age of 274.8 ± 1.5 Ma from the Collingham Formation of the Laingsburg depocentre, 50 km to the southeast (Fildani *et al.* 2007; 20.75°E). The

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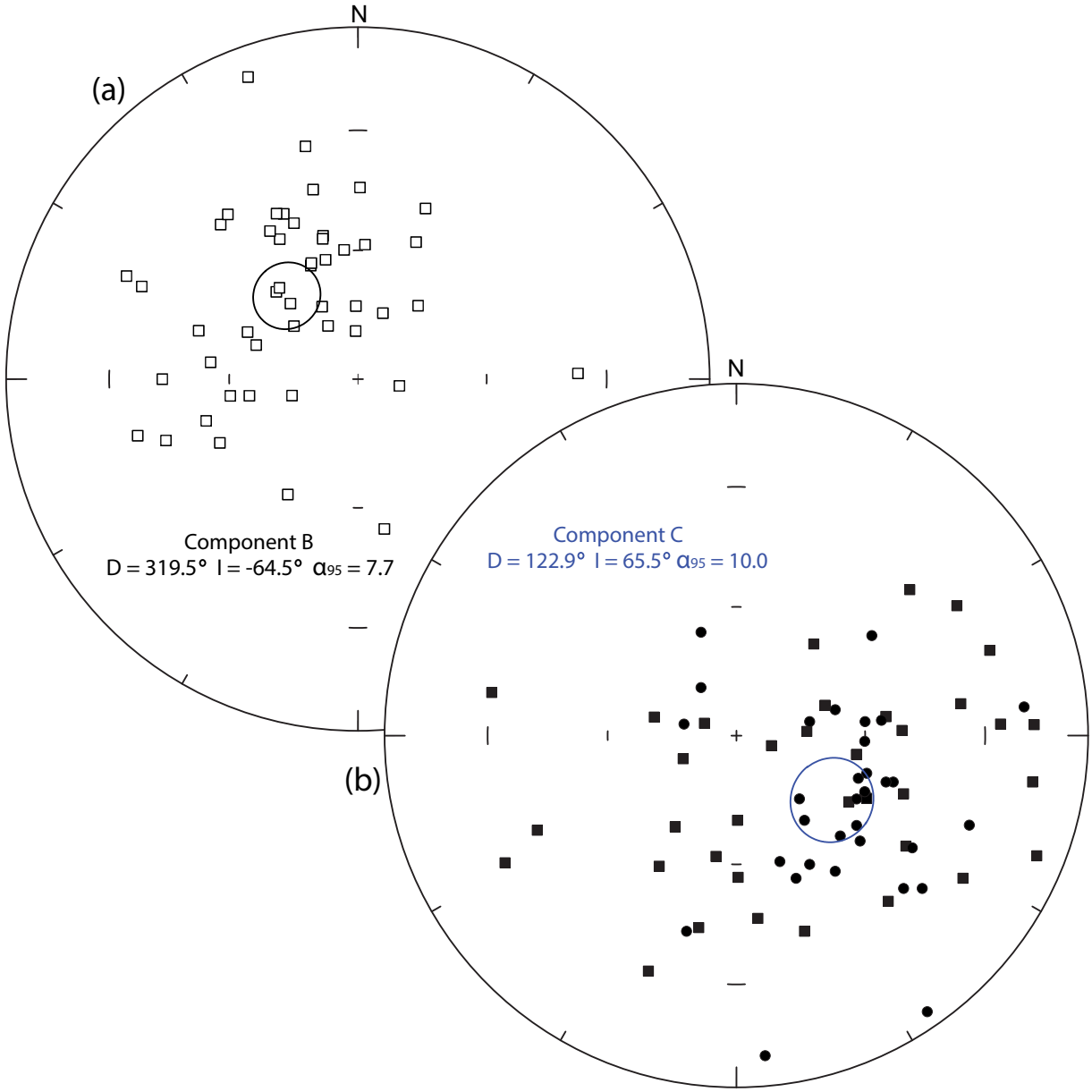


Figure 11. Equal area diagrams displaying the (a) mean overprint direction (Component B) and the (b) reverse ChRM direction (Component C) of the upper Waterford Formation at SL1. Up (down) pointing palaeomagnetic directions are indicated by open (closed) squares in the stereoplots. Arc constraints for samples that do not reach stable endpoints are calculated with the method of McFadden & McElhinny (1988), and are shown with circle symbols. Black and blue ovals represent the cone of 95 per cent confidence about the mean direction.

magnetostratigraphy and corresponding ages, therefore, support an age range of *ca.* 290–265 Ma for the deposition of the Ecca Group of the southern Karoo Basin. This age range is consistent with the accepted depositional history for the Karoo Basin, as well as the large body of published biostratigraphic data, and re-establishes the location of the geochronologically constrained PTB of the Tanqua depocentre to within the overlying terrestrial sediments of the Beaufort Group (Fig. 15).

5.2 SHRIMP U–Pb geochronology

Samples OR72 and OR98 reported here from the Ecca Group are plotted against the corresponding sampling level for comparison with the magnetostratigraphy (Fig. 13). The U–Pb results from OR72 indicate that the system has likely been exposed to some level of Pb-loss (Fig. 13). Similar to other published isotopic analyses from the Ecca Group, we isolated several late Permian-aged grains from each sample (Table 3). These grains formed a distinct subpopulation in OR98, but were not excluded from the final age calculation in order to provide a minimum age constraint (Fig. 13). The age reported here is *ca.* 10 Ma older than those previously reported from the Ecca Group (Fig. 2), and we interpret this as mainly resulting from the contrasting interpretations of complicated U–Pb data sets. (Fildani *et al.* 2007, 2009; McKay *et al.* 2015, 2016). The former age calculations were based on the single youngest grain (Fildani *et al.* 2007, 2009) or subpopulation of youngest grains (McKay *et al.* 2015, 2016), whereas our analysis was based on the weighted mean $^{206}Pb/^{238}U$ age of all isolated Permian grains (<280 Ma) that met the specified quality criteria (Section 3.3).

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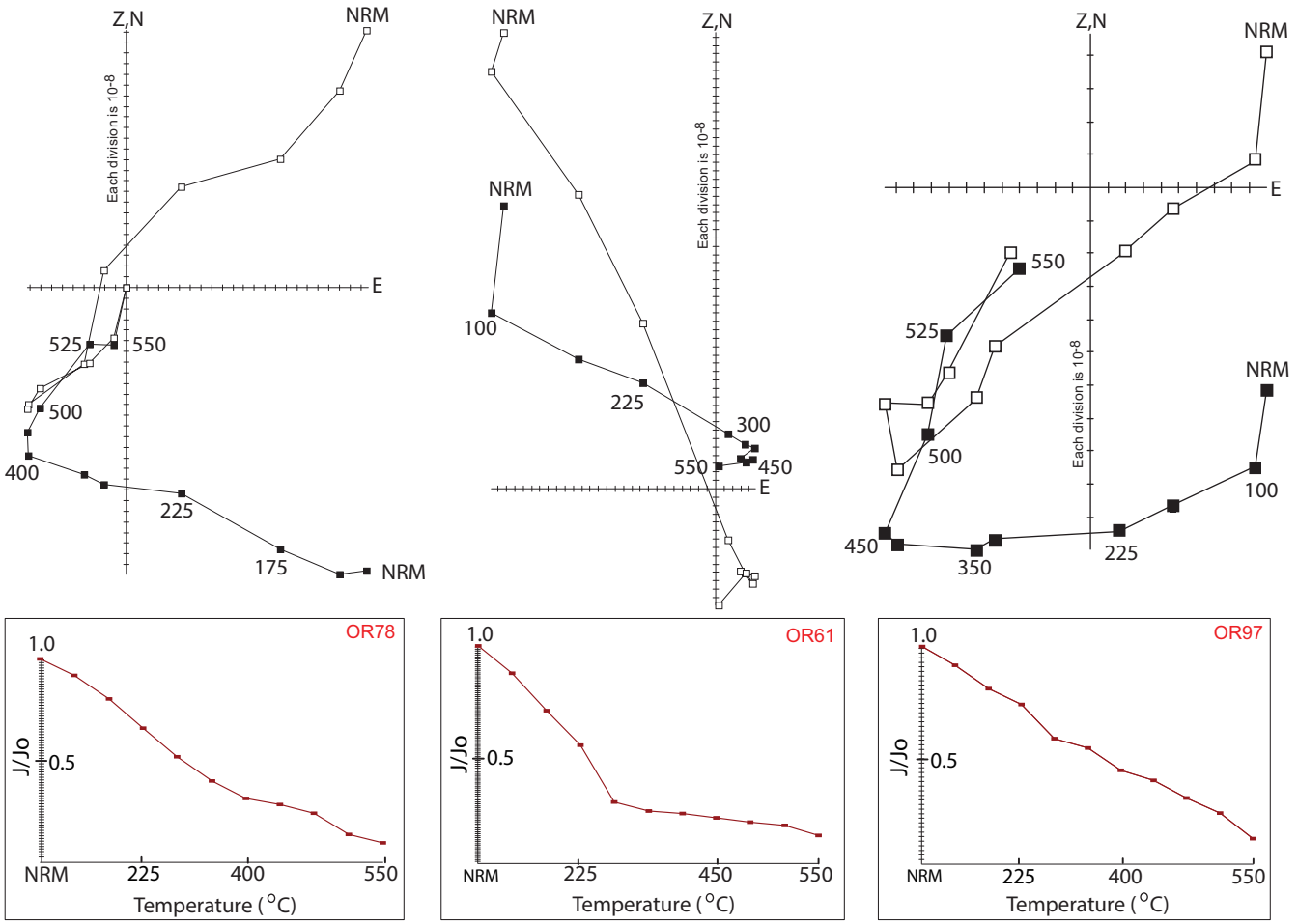


Figure 12. Representative Zijderveld and intensity decay diagrams for the Skoorsteenberg Formation from the OR1 core (vertical orientation data). The Jurassic overprint direction is removed by 400 °C, while the reverse polarity ChRM is isolated from 450 to 550 °C. Demagnetization curves are indicative of magnetite as the main magnetic phase. Solid (open) squares represent projections on the horizontal (vertical) plane.

We identified several inherited grains with ages of 1130, 880, 590, 450 and 380 Ma (Table 3). The grains selected for our age calculation showed no obvious signs of inheritance, and SHRIMP spots were taken on rims to avoid contamination from potential inherited cores (Fig. 5b).

Interestingly, the work by McKay *et al.* (2015, 2016) did not identify any ages older than *ca.* 280 Ma in the Ecca Group, and so early Permian-aged zircons were excluded from their age calculations by reasoning of assumed inheritance. This assumption was based on the hypothesis that the zircons from the overlying Beaufort Group (*ca.* 262–268 Ma) were sourced from the same volcanic centre as the Choiyoi Group of South America, that they proposed to have experienced a period of Triassic zircon exhaustion and recycling of older inherited grains (McKay *et al.* 2015). However, recent rare Earth element (REE) data indicate that the Karoo Basin ash beds were more likely sourced from Antarctica, and do not share a common magmatic history with the South American ash beds (McKay *et al.* 2016). Without clear documentation of the location of the magma chamber from which the ashes were generated, applying a hypothetical inheritance cut-off to the total analyses would skew the calculation towards a younger age.

Here, we provide a more straightforward explanation for this apparent age inversion: that the zircon grains from ash beds in the middle Ecca Group have been affected by Pb-loss, the degree of which is difficult to assess at the resolution of the SHRIMP method for samples <500 Ma (Tohver *et al.* 2015). For example, for Phanerozoic samples, subsequent loss of radiogenic Pb will cause analyses to plot in an array between the magmatic age and the age of disturbance along a line subparallel to the Concordia curve, and with uncertainty ellipses still overlapping with Concordia, so the degree of Pb-loss is potentially impossible to resolve (Tohver *et al.* 2015). The reproducibility of consistently younger ages from the underlying Ecca Group and older ages from the overlying Beaufort Group, along the southern margin of the Karoo Basin, can be explained by an episode of Pb-loss. Our palaeomagnetic analysis from the Tierberg and Skoorsteenberg Formations (OR1 core) shows a much larger degree of remagnetization with generally more unstable demagnetization behaviour than in the overlying Kookfontein and Waterford Formations, suggesting that the lower to middle Ecca Group was more heavily affected by this Pb-loss event than the upper Ecca Group, though this may also be a result of the drill string.

Some published illite crystallinity data from the middle Ecca Group support low-grade metamorphic temperatures of 170–200 °C (Hälbich *et al.* 1983), which is within range of when fluid-mediated lattice damage initiates in zircon (Geisler *et al.* 2002, 2007). However, we did not find any clear evidence for re-absorption or alteration of the oscillatory zoning in the zircons (Figs 5a and b), and the measured $^{232}Th/^{238}U$ ratios are not consistent with a lattice disturbed by metamorphism (Table 3). The CL and BSE images of the analysed zircon grains (Fig. 5) do not show extensive U

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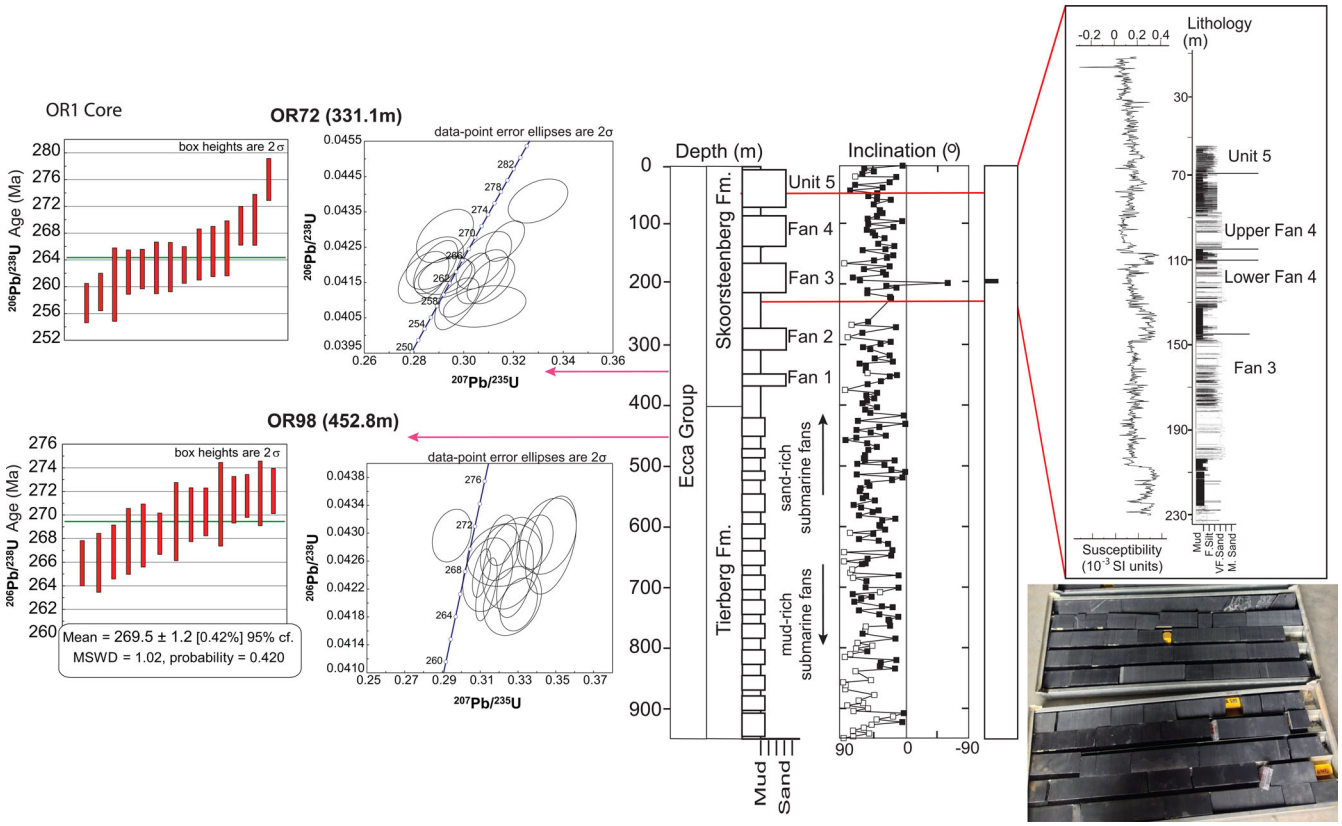


Figure 13. Magnetostratigraphy of the Skoorsteenberg and Tierberg Formations from the OR1 core (see Fig. 1 for location). ChRM inclination (polarity) data are plotted against sampling level, U–Pb SHRIMP ages, lithology (Hodgson *et al.* 2011) and magnetic susceptibility measurements (SI units). Closed (open) squares represent directions isolated using lines (arcs). The section is restricted to reverse polarity (white), with one potential normal subchron (black). The age presented here is a weighted mean $^{206}Pb/^{238}U$ age. Corresponding Wetherill diagrams are shown only for reference. Pink arrows indicate the stratigraphic levels of the dated ash beds.

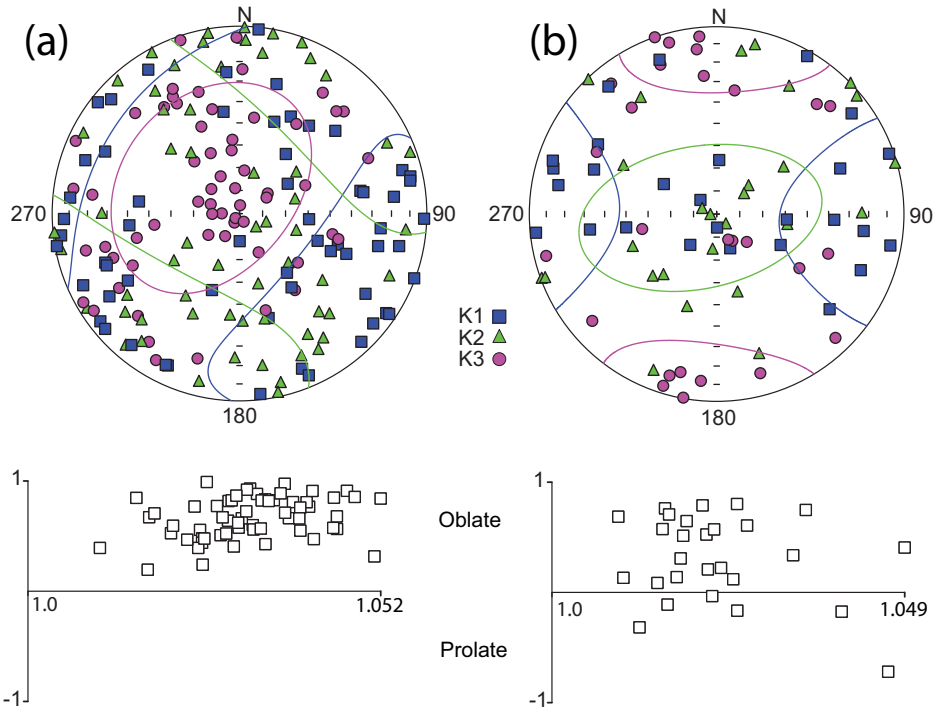


Figure 14. Principal axes of magnetic susceptibility for the Ecca Group at (a) Pienaarsfontein and (b) SL1. Top: equal-area plots of the directions of the principal axes of AMS with $K1 = K_{max}$, $K2 = K_{int}$, $K3 = K_{min}$. Ovals represent the confidence ellipses for the mean direction. Bottom: Jelinek diagrams displaying the general shape of the AMS ellipsoids with P = degree of anisotropy and T = shape factor.

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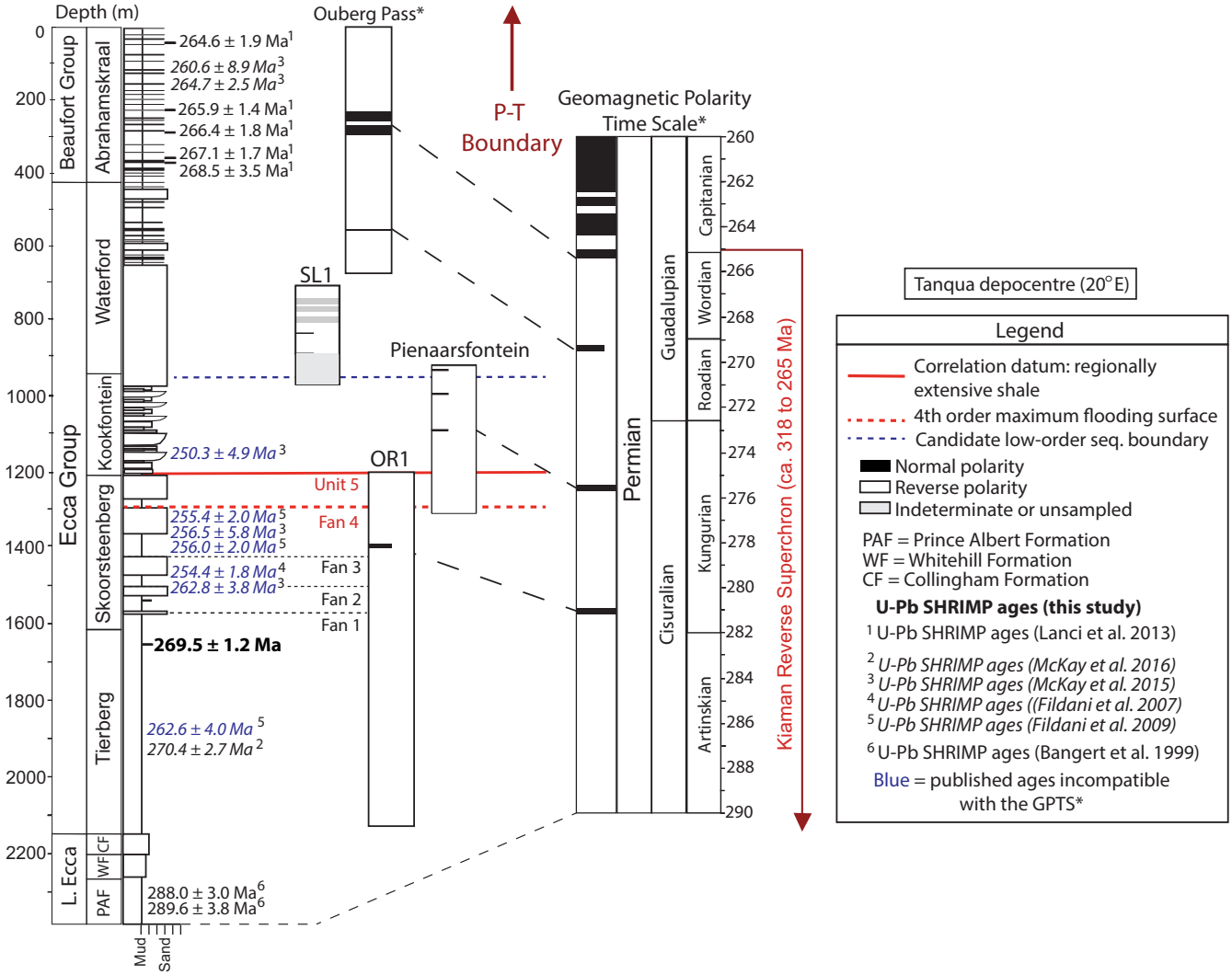


Figure 15. Lithostratigraphy, corresponding U–Pb ages, and magnetostratigraphic composite of the Permian Ecca Group and Lower Abrahamskraal Formation (Beaufort Group) of the Tanqua depocentre. GPTS* after Hounslow & Balabanov (2016) with geological stage boundaries after Ogg *et al.* (2016) and lithology after Hodgson *et al.* (2011). Published ages shown in blue are incompatible with the reported magnetostratigraphy. *Ouberg Pass = Lanci *et al.* (2013).

damage, metamorphism or any other feature that indicates zircon damage, so Pb-loss is a reasonable explanation to the spread of ages observed in the data collected. The Pb-loss hypothesis stems from the observation that the previously published SHRIMP ages from the Ecca Group are consistently too young when compared to the overlying Beaufort Group (Fildani *et al.* 2007, 2009; McKay *et al.* 2015, 2016), and if their inaccuracy came from heterogeneous standards or other analytical errors, the age difference would be non-systematic.

As this Pb-loss event has only been observed so far in the Ecca Group, we conclude that this episode was most likely an isolated incident. However, another possibility is that this Pb-loss event was more extensive, and the preponderance of younger grains isolated from the Ecca Group could be explained by a lower proportion of co-eruptive zircon (McKay *et al.* 2015) than in the overlying Beaufort Group. Zircon has been shown to be present in variable concentrations in different tectonostratigraphic terranes (Moecher & Samson 2006). This variability also has the potential to create an age bias in the study of magmatic systems (McKay *et al.* 2015). For example, the Beaufort Group may have been sourced by zircon-rich volcanic eruptions, while the Ecca Group was sourced by zircon-poor eruptions. In this scenario, the Beaufort Group would have been exposed to the same Pb-loss event; however, the abundance of available zircon grains would dilute these disturbed analyses in the final age calculation, since they would not fit within the main population (or subpopulation), but represent outliers. Interestingly, Pb-loss was similarly detected in the Transantarctic Mountains of Antarctica, where a large number of Triassic grains were present in the interpreted Permian-aged beds (Elliot *et al.* 2017). As REE data seem to support a common magmatic origin for the Karoo Basin and Antarctic ash beds (McKay *et al.* 2016), further research in these areas is required. Regardless of a direct demonstration or mechanism for the proposed Pb-loss, the magnetostratigraphy presented here provides an independent test against potentially disturbed U–Pb isotopic data, and confirms the Early Permian age for the Ecca Group.

5.3 The Gondwana APWP

In order to further constrain the relative timing of the interpreted primary and secondary magnetizations, the mean poles reported from Component B (overprint) and Component C (ChRM) are

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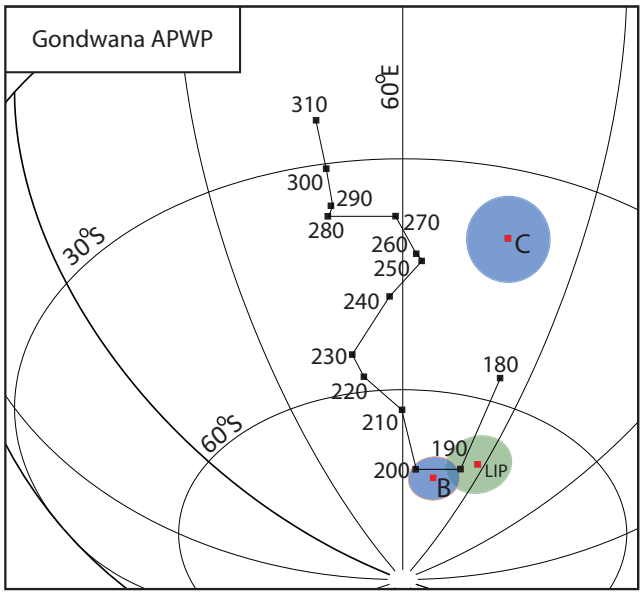


Figure 16. Components B (overprint) and C (ChRM) plotted alongside the recommended Gondwana APWP of Torsvik *et al.* (2012) in an orthogonal projection centred at 40°S, 60°E. Blue = this study; Green = Palaeomagnetic results from the Karoo LIP at *ca.*183 Ma (Moulin *et al.* 2017).

plotted alongside the recommended APWP of Torsvik *et al.* (2012) for reference (Fig. 16). The reported ChRM direction is consistent with the 270 Ma segment of the Gondwana APWP, and forms a critical angle (λ_c) of 13.6° with the reported *ca.* 270 Ma RM (running mean) Gondwana pole (no applied corrections; Torsvik *et al.* 2012). Component B is consistent with the 190 Ma segment ($\lambda_c = 8.3°$) of the Gondwana APWP, as well as recent results reported from the Karoo LIP (Moulin *et al.* 2017). This suggests that this component represents a partial (low-temperature) Jurassic remagnetization of the Ecca Group during the initial stages of LIP emplacement. The eruption of the Karoo LIP likely resulted in the mobilization of heated fluids which could leach Pb, and recent palaeothermal studies of the basin have indicated regional elevation of palaeotemperatures of the sedimentary rocks of the Ecca Group to 200 °C (Maré *et al.* 2016). This temperature interval is consistent with potential Pb-loss, which should not occur in zircons except at low temperatures (Schoene 2014), such as during low-temperature hydrothermal dissolution and reprecipitation (Geisler *et al.* 2002, 2003).

Though Karoo Basin palaeomagnetic directions were originally considered to be representative of a complete remagnetization (demagnetization to 350 °C) due to a negative fold test in the Cape Fold Belt (Ballard *et al.* 1986), a number of palaeomagnetic studies have since been published which resolved primary magnetization components suitable for comparison with the GPTS (De Kock & Kirschvink 2004; Ward *et al.* 2005; Lanci *et al.* 2013; Gastaldo *et al.* 2015; Tohver *et al.* 2015). The Cape Orogeny (*ca.* 275–265 Ma), though significant in deformation in some areas, does not seem to have significantly remagnetized the Ecca Group strata, which in this study location (Tanqua depocentre; Fig. 1) are almost flat-lying and unaltered. Palaeomagnetic results from a stratigraphically overlying section at Ouberg Pass (Lanci *et al.* 2013) revealed magnetizations with several antipodal reversed and normal polarities that would be impossible to obtain if the remanence represented a blanket remagnetization acquired during KRS-aged Cape Fold Belt activity. However, if the magnetizations were reset during the KRS, then this would imply an even older age (>270 Ma) for the Ecca Group, consistent with the proposed hypothesis of Pb-loss and the main conclusions of this work. If the Cape Orogeny had caused the Ecca Group Pb-loss, then it had to have occurred directly after the deposition of the Ecca Group and before the deposition of the Beaufort Group. Hence, it would be more or less coeval with sediment deposition. As the Ecca Group is outside the main area of Cape Fold Belt deformation (Fig. 1), we attribute the thermal overprint marker to the eruption of the Karoo LIP, which due to the vicinity of numerous dykes and sills in the field area, had the potential to make a more significant impact on resetting the magnetizations than distal Cape Fold Belt activity.

6 CONCLUSIONS

In an attempt to better resolve the Permian chronostratigraphy of the Karoo Basin, we have presented magnetostratigraphic and geochronologic data from a large part of the Ecca Group in the Tanqua depocentre. A uniformly reverse polarity magnetostratigraphic profile of a composite ~1500 m section confirms that deposition occurred during the KRS (*ca.* 318–265 Ma), providing a minimum depositional age constraint of *ca.* 265 Ma for the upper Ecca Group. This age is not compatible with the geochronologically indicated PTB in the upper Skoorsteenberg Formation, but supports the existing biostratigraphic data in the overlying Beaufort Group. A U–Pb age of 269.5 ± 1.2 Ma is reported here from the uppermost Tierberg Formation, and is interpreted as a minimum age constraint due to a proposed Pb-loss event in the Ecca Group of the southern Karoo Basin.

A combination of published U–Pb ages from the overlying Abrahamskraal Formation (*ca.* 262–268 Ma), the Prince Albert Formation (*ca.* 290 Ma) and the Collingham Formation (*ca.* 275 Ma), compared with our magnetostratigraphic composite of the Tanqua depocentre, support an age range of *ca.* 290–265 Ma for the ~1.5 km thick Skoorsteenberg/Kookfontein/Waterford Formation section of the Ecca Group.

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